Paediatric Urology 1

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · **P2** = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 41	Management Strategies for Vesicoureteral Reflux
Ch 49	Anomalies of the Kidney and Collecting System
Ch 50	Renal Dysgenesis and Cystic Disease of the Kidney
Ch 51	Pathophysiology of Urinary Tract Obstruction
Ch 52	Ectopic Ureter, Ureterocele and Ureteral Anomalies
Ch 53	Surgery of the Ureter in Children
Ch 56	Hypospadias
Ch 57	Etiology, Diagnosis and Management of the Undescended Testis
Ch 61	Urologic Considerations in Pediatric Renal Transplantation
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Ch 64	Paediatric Urological Oncology: Pelvis, Bladder, Testis and Genitalia
★	Putting it together: common paediatric decisions

41 Management Strategies for Vesicoureteral Reflux

1 · DEFINITION AND CONCEPT

We treat the infection risk, not the reflux.

- **Vesicoureteral reflux (VUR)** is retrograde urine flow to the upper tract, linked to **acquired scarring** and **congenital dysplasia**.
- **Infection, not reflux, is the injury**, so aim for sterile urine and good emptying.

2 · EMBRYOLOGY AND ANTIREFLUX MECHANISM

P1 **P4** *Primary (congenital) VUR: VUJ embryology, pathophysiology, natural history and management — Mackie–Stephens bud position (early bud gives high lateral orifice, VUR and dysplasia; late bud gives ectopia).*

*Passive flap valve; Paquin tunnel length to diameter **5:1** vs 1.4:1 in refluxing units.*

Not in the paper answer:

- Abnormal bud position also causes hypoplasia, so some "scars" are congenital.

3 · PRIMARY VS SECONDARY VUR

Secondary reflux needs the bladder treated first.

P4 *VUR: etiopathogenesis, recent trends in management and STING technique — primary (deficient UVJ) vs secondary (PUV, neurogenic, dysfunctional voiding, ureterocele, duplication, diverticulum); overlap; detrusor overactivity commonest urodynamic finding; infant male high voiding pressures*

Not in the paper answer:

- In spina bifida, VUR occurs in up to **30%**: start early clean intermittent catheterisation (CIC) and an anticholinergic or β3-agonist.
- **Reflux at low pressure or in early filling means poor resolution.** Treating detrusor overactivity improves resolution.

> **KEY POINT**
> Primary and secondary are not exclusive: "primary" VUR often has a treatable secondary component (overactivity, bladder and bowel dysfunction). Remove it and only true primary VUR remains.

4 · EPIDEMIOLOGY

- VUR is found in **~30%** of children with UTI and ~17% of those imaged for hydronephrosis. It is present in **38.6%** under 2 years versus 7.6% at 12–21 years.
- Girls are affected ~2× overall, but infant boys predominate.

5 · GRADING — INTERNATIONAL REFLUX STUDY (1981)

Grade	Ureter	Pelvis and calyces
I	Non-dilated, ureter only	Not reached
II	Not dilated	Reached, no dilatation
III	Mild–moderate dilatation	Mild–moderate dilatation, minimal forniceal blunting
IV	Moderate tortuosity	Moderate dilatation
V	Gross dilatation and tortuosity	Gross dilatation; **papillary impressions lost**

- Grades I–II are non-dilating and III–V dilating. Radionuclide cystography (RNC) uses 3 grades.
- **Ureteral diameter ratio (UDR)**, the largest distal ureteric diameter divided by the L1–L3 vertebral distance, predicts resolution and breakthrough UTI (BUTI) better than grade.

6 · DIAGNOSIS

When to image after UTI

- **AAP 2011**, febrile UTI at 2–24 months: do **ultrasound (US) first**, and a voiding cystourethrogram (VCUG) only if US is abnormal or after a **second febrile UTI**.
- **NICE**: US, with VCUG mainly for atypical or recurrent UTI in younger children.
- **Top-down**: DMSA scan first, VCUG only if abnormal. A normal DMSA leaves a **<1%** chance of high-grade VUR but misses ~1/3 of (mostly low-grade) VUR.

Cystographic methods

P1 *VCUG — gold standard; gravity ≤70 cm H_2O; early study once urine sterile; cyclic VCUG +12–20%; filling-phase reflux poor prognosis; postvoid pelvic retention suggests UPJO; RNC and ceVUS alternatives*

Not in the paper answer:

- Direct RNC is more sensitive for grades II–V but **misses grade I**.
- Suspect coexisting **UVJ obstruction** with a distal aperistaltic segment, dilute contrast and delayed drainage. It is a **relative contraindication to endoscopic injection**.

Upper tract

- **Renal and bladder ultrasound (RBUS)** is the serial mainstay but cannot exclude VUR.
- **DMSA** is the gold standard for cortical defects (sensitivity 98%, specificity 92%); a defect that persists is a scar.

7 · CONGENITAL DYSPLASIA VS ACQUIRED SCAR

Only the acquired scar is preventable.

P4 **P1** *VUR: etiopathogenesis, recent trends in management and STING technique — dysplasia (boy, IV–V, antenatal, not preventable) vs acquired scar (girl, febrile UTI, BBD, preventable); polar compound papillae, intrarenal reflux at 2 mmHg; Ransley–Risdon sterile reflux; "big bang" debated*

Not in the paper answer:

- Hypertension develops in **34–38%** of scarred children. **Correcting VUR does not lower blood pressure**; removing a small kidney or segment may.

8 · BLADDER AND BOWEL DYSFUNCTION (BBD)

BBD is the main modifiable factor.

- **Treat constipation first**; it causes sphincter co-contraction.
- BBD brings more UTI on prophylaxis and after surgery, less resolution, and **lower endoscopic success, but not open surgical success** (AUA).
- **Phimosis**: in boys with VUR, circumcision cuts the number needed to treat (NNT) to prevent one UTI from ~111 to **~4**. Topical steroid is equivalent.

9 · NATURAL HISTORY AND RESOLUTION

Grade	Spontaneous resolution
I–II	Most; grade II 63–85%
III	~50%; ~90% of resolutions within 4 years
IV–V, bilateral III	Few (infants with IV–V resolve well in year 1)

- Resolution favours **low grade, young age, unilateral reflux, no BBD, voiding-only reflux, higher bladder volume at onset and low UDR**.

10 · MEDICAL MANAGEMENT — CONTINUOUS ANTIBIOTIC PROPHYLAXIS (CAP)

CAP failure is the main trigger for surgery.

P3 *Management of bilateral VUR: medical treatment, surgical techniques, selection and long-term outcome — CAP agents by age with doses (TMP/amoxicillin <2 months; TMP-SMX, nitrofurantoin >2 months).*

Hidas risk groups, indications for correction, 8–10% febrile UTI after stopping CAP.

Not in the paper answer:

- Give a low dose **at night**; avoid TMP-SMX in neonates (bilirubin displacement).
- **If febrile UTI breaks through CAP despite good bladder and bowel habits, stop medical management and correct the VUR.**

Risk stratification (Hidas)

Factor	Low risk	High risk
Sex / age	Male; >5 y	Female; <1 y
Laterality / grade	Unilateral; I–II	Bilateral; III–V
BBD / scars	No	Yes
Presentation	Prenatal hydronephrosis	Multiple febrile UTIs

- Two-year BUTI on CAP is **8.6% low, 26% intermediate, 62.5% high** risk. About two-thirds of children are low risk.

11 · PRACTICAL ALGORITHM (AUA-BASED, RISK-STRATIFIED)

ALGORITHM

1. Confirm true febrile UTIs; do RBUS. If multiple febrile UTIs or upper-tract findings, then VCUG.
2. In all, treat BBD (constipation first) with timed voiding.
3. If low risk (grade I–II, no BBD, male, older), then observe with or without CAP and treat febrile UTI promptly.
4. If high risk (grade ≥III, girl, BBD, scars, infant), then give CAP.
5. If breakthrough febrile UTI, new scars, or persistent high-grade VUR, then correct (endoscopic or reimplant).

- CAP helps little at grade ≤II without BBD, but helps at ≥III, at least in girls.

12 · SURGICAL CORRECTION

Surgery stops reflux but does not protect kidneys better.

- Open correction succeeds in **>95%**, but neither open nor robotic surgery beats medical therapy for scarring or growth. The adult UTI risk persists.
- **Endoscopic dextranomer/hyaluronic acid (Dx/HA)**: BBD lowers success; avoid it with UVJ obstruction. Calcified mounds can **mimic ureteric stones**.
- After a cross-trigonal reimplant (Ch 53), **retrograde ureteric access is difficult**.

13 · ASSOCIATED CONDITIONS

- **UPJO with VUR**: **correct the obstruction first**; repairing both at once risks the ureteric blood supply.
- In a **duplex** kidney, VUR usually enters the **lower pole** (Weigert–Meyer).
- A **ureter entering a paraureteral diverticulum** will not resolve; a diverticulum beside the orifice resolves like primary VUR.
- Contralateral VUR with a multicystic dysplastic kidney usually resolves.
- **Megacystis–megaureter** (massive VUR recycling urine into a large thin bladder) and **obstructive refluxing megaureter** are treated by reimplant, not injection or outlet surgery.

14 · SIBLING SCREENING

- Sibling prevalence is ~**30%**, higher when younger.
- The target is **renal damage, not VUR**. AUA: **VCUG for siblings with a cortical abnormality or renal asymmetry on US, or a UTI history**; otherwise observe.

15 · FOLLOW-UP AND ADULTHOOD

- If scarred, check BP, proteinuria and renal function for life.
- **Pregnancy**: scars bring ~3× hypertension and **~8× pre-eclampsia**. **Scarring, not VUR, drives morbidity.**

16 · LANDMARK TRIALS

Trial	Design	Result	Take-home
IRSC	<9 y, high grade; CAP vs open surgery	UTI equal (38%); surgery fewer pyelonephritis; scars, 10-y function equal	Surgery ≈ medical for scarring
Birmingham Reflux Study	High grade; medical vs surgical, 5 y	New scars equal, all within 2 y	Equal; supports "big bang"

Trial	Design	Result	Take-home
Swedish Reflux Trial	1–2 y, grade III–IV; CAP vs endoscopic vs surveillance	Girls febrile UTI: 19% CAP, 23% endoscopic, **57%** surveillance; boys no difference	Treat girls with dilating VUR
RIVUR	607 children, grade I–IV after first/second UTI; TMP-SMX vs placebo	Recurrent UTI **12.9% vs 23.6%**; scarring unchanged	CAP prevents UTI, not scars
PRIVENT	576 children with UTI; CAP vs placebo	Absolute UTI reduction 6% (19% to 13%)	Modest benefit
French / low-risk trials	Grade I–III; CAP vs none	No significant difference	No benefit at low risk

KEY POINT

Trials disagree because of population risk: RIVUR was diluted by low-risk children; reanalysis gives NNT 5 high-risk vs 18 low-risk. Neither CAP nor surgery prevents scarring better.

49 Anomalies of the Kidney and Collecting System

1 · EMBRYOLOGY THAT EXPLAINS EVERYTHING

- The ureteric bud (UB) arises from the caudal wolffian duct (WD) at **5–7 wk**, invades the metanephric mesenchyme (MM) and branches; its tips induce nephrons (**reciprocal induction**).
- In weeks 6–9 the kidney ascends and rotates 90° ventromedially, so the pelvis faces medially.
- Renal anomalies travel with WD (male) and müllerian duct (MD, female) anomalies; the adrenal develops independently.

KEY POINT

Every anomaly of number is a failure of UB–MM induction: timing and extent set the phenotype. An early insult drags the WD/MD with it (genital anomalies). Arrested ascent leaves an unrotated kidney with anterior pelvis and fetal blood supply; fusion freezes kidneys where they meet.

2 · BILATERAL RENAL AGENESIS (BRA)

BRA is uniformly lethal, so recognise it and counsel.

- It occurs in **≈1 in 5000** births, **75% male**. First-degree relatives carry ≈1000× risk, so **scan parents and siblings by ultrasound (US)** of any infant with unilateral agenesis (URA), BRA or dysgenesis.
- The ureters are atretic and the bladder absent or hypoplastic. The flat adrenal gives the **"lying down" sign**.
- **Potter facies** and limb deformities are **deformations** from absent amniotic fluid, not malformations. Undescended testes (43%) and gastrointestinal anomalies (≈60%) are common.
- **Pulmonary hypoplasia** is the cause of death. **Peters two-step model**: a kidney-derived "renal factor" acts before 16 wk, then amniotic fluid in the canalicular phase (16–26 wk), so fluid alone only partly restores the lungs.
- **Diagnosis**: prenatally, severe oligohydramnios, no kidneys and a small chest. In a neonate, **anuria >24 h without bladder distension** means BRA until proved otherwise; confirm with US and power Doppler.
- **Prognosis**: ≈40% are stillborn and the rest die in 24–48 h. Amnioinfusion is experimental (RAFT, §9) and needs ethics oversight with full counselling.

KEY POINT

BRA is lethal through the lungs, not the kidneys. Amnioinfusion replaces only the amniotic step, so it converts pulmonary deaths into neonatal end-stage renal disease (ESRD).

3 · UNILATERAL RENAL AGENESIS (URA) AND THE SOLITARY FUNCTIONING KIDNEY (SFK)

The issues are genital anomalies and lifelong hyperfiltration.

- URA occurs in **1 in 1200**, male 1.8:1. Many apparent URAs are an **involuted multicystic dysplastic kidney (MCDK)**.
- Associations include VACTERL, MURCS, Turner, Kallmann, branchio-oto-renal and DiGeorge syndromes. **Maternal diabetes triples the risk**. An isolated preauricular pit or single umbilical artery needs no US; two or more anomalies do.
- **The ipsilateral ureter is absent in ≈60%**. The defect is in the UB, so the gonad is rarely affected.
- **Contralateral kidney**: vesicoureteral reflux (VUR) **≈30%**, pelviureteric junction (PUJ) obstruction ≈11%, vesicoureteric junction (VUJ) obstruction ≈7%.
- **Genital anomalies occur in 20–40%** (females 25–50%, males 10–15%).

	Male (WD derivatives)	Female (MD derivatives)
Always present	Testis, head of epididymis	Fimbriated end of tube
Commonest	Absent vas/epididymal body-tail/seminal vesicle (≈50%). **79% of men with absent vas lack the ipsilateral kidney**	Unicornuate/bicornuate, didelphys or septate uterus; vaginal atresia/duplication; Gartner duct cyst with blind ectopic ureter
Named syndrome	**Zinner**: seminal vesicle cyst and ipsilateral agenesis	**OHVIRA** (obstructed hemivagina and ipsilateral renal anomaly; Herlyn–Werner–Wunderlich): pubertal haematocolpos, pain
Trigger to image kidneys	Impalpable vas (infertility, herniotomy/orchidopexy)	Any uterine anomaly; pelvic US after thelarche, before menarche

- **Magee classification**: type I (before wk 4) gives a unicornuate uterus; type II (early wk 4) gives didelphys with obstructed hemivagina on the agenesis side; type III (after wk 4) gives isolated URA with normal genitalia.
- **MRKH** (46,XX müllerian agenesis): **type II has renal anomalies in 40–60%**; type I rarely does.
- **Work-up**: compensatory hypertrophy is seen from 20 wk in ≈90%. DMSA finds a missed ectopic kidney. A voiding cystourethrogram (VCUG) for contralateral VUR is **not routine**. Sport is not precluded.

Prognosis and surveillance of the SFK

- **Brenner hyperfiltration**: reduced nephron mass leads to glomerular hypertension, then albuminuria and sclerosis. Risk is highest when the SFK itself has a congenital anomaly (CAKUT), low birth weight or prematurity.
- In KIMONO, ≈⅓ had renal injury by ≈10 y, so follow-up is lifelong.

Situation	Blood pressure (BP) and first-morning urine albumin	Ultrasound	Creatinine
SFK without CAKUT	Annually	Every 5 y if glomerular filtration rate (GFR) >60 and no medication	Periodic, into adolescence
SFK with CAKUT	Twice yearly	As indicated	Periodic

Situation	Blood pressure (BP) and first-morning urine albumin	Ultrasound	Creatinine
GFR <60 or on antihypertensive/antiproteinuric	BP, albumin, creatinine 2–4×/year		

Hypertension or microalbuminuria (>30 mg/24 h) calls for an ACE inhibitor, salt limitation and no excess protein.

4 · SUPERNUMERARY KIDNEY

- It has its **own capsule, collecting system and separate blood supply**, unlike a duplex kidney. About half are hydronephrotic, with obstruction, infection or stones in early adulthood.

5 · ECTOPIA (ANOMALIES OF ASCENT)

An ectopic kidney is easily mistaken for agenesis or tumour.

Simple ectopia

- It occurs in **1 in 900** at autopsy and is bilateral in ≈10%. **Pelvic is commonest**.
- The kidney is small and malrotated (pelvis anterior). **Hydronephrosis is common**, half from obstruction (mostly PUJ). **VUR occurs in 30–50%**, often into the orthotopic kidney.
- The blood supply is anomalous: **map it by CT or MR angiography (MRA) before surgery**.
- Genital anomalies occur in 15–45%. Continuous incontinence in a girl suggests an ectopic ureter from the ectopic kidney.
- **Never remove a "pelvic mass" without imaging**: solitary pelvic kidneys have been excised as tumours.

Cephalad and thoracic kidney

- Cephalad ectopia is seen with **omphalocele**. A thoracic kidney lies in the **foramen of Bochdalek**, is usually silent, and needs CT or MRU rather than thoracotomy.

6 · FUSION ANOMALIES

Fused kidneys bring obstruction, stones and variable vessels.

Crossed renal ectopia

- The kidney lies opposite its ureteric orifice; **90% are fused**, making it the second commonest fusion anomaly. Left-to-right crossing is 3:1, and the ureteric orifice is orthotopic.
- **McDonald–McClellan types**: with fusion (90%), without fusion, solitary crossed and bilateral crossed (rarest).

Crossed fused form	Key feature
Inferior ectopia (≈⅔, commonest)	Crossed upper pole fused to normal lower pole; pelves anterior
Sigmoid (S-shaped)	Second commonest; pelves face opposite ways
Lump (cake)	Lobulated pelvic mass
L-shaped (tandem)	Crossed kidney transverse, anterior to L4
Disc (pancake)	Medial fusion, reniform outline kept
Superior ectopia	Least common; crossed lower pole to normal upper pole

- An **anterior pelvis means early fusion**; a medial pelvis means late fusion. VUR occurs in ≈20%.
- **A stone in the crossed ureter hurts on the anephric side**; pyelonephritis or PUJ obstruction hurts where the kidneys lie. Obtain MRU or MRA before major surgery.

Horseshoe kidney

- It is the **commonest fusion anomaly**: **1 in 400** (0.25%), male 2:1. **Lower-pole fusion occurs in 95%**, before rotation.
- The **isthmus** lies at L3/L4, **just below the IMA**, in front of the aorta and IVC.
- The calyces **point posteriorly**, and the ureter inserts high and bends over the isthmus. **The blood supply is highly variable**, so obtain MRA or CT before surgery.
- **≈60% of Turner patients have a horseshoe kidney**; it is also associated with neural tube, anorectal and VATER anomalies.
- **PUJ obstruction occurs in 13–34%**, but many hydronephroses are not obstructed. VUR occurs in 8–32%.
- **Stones affect ≈⅓ of adults**, so still do a metabolic work-up.
- **Wilms risk is 1.76–7.9×**; preoperative chemotherapy preserved ≈75% of parenchyma (NWTS). Renal cell carcinoma is not increased.
- At least 50% are asymptomatic. Pregnancy is unaffected and the kidney is transplantable.

> **KEY POINT**
> † PUJ obstruction in horseshoe kidney: dismembered pyeloplasty with dependent anterior anastomosis; symphysiotomy not routinely needed.

7 · MALROTATION AND VASCULAR ANOMALIES

- Malrotation occurs in ≈1 in 939. **Weyrauch** types: **ventral (no rotation, commonest)**, ventromedial, dorsal (rarest) and lateral (reverse). **The vessel course is the only clue** to direction.
- Hilar fibrosis can cause intermittent PUJ obstruction and stones; suspect malrotation when a stone lies in an unusual position.

P2 *Anomalies of renal artery. — multiple / aberrant / accessory terms; single artery 71–85%; end-arteries; crossing vessels in normal PUJs; aneurysm (Abeshouse, treat >2 cm, pregnancy, symptoms). Congenital vs acquired AVF; embolisation.*

Not in the paper answer:

- Treat an aneurysm for rupture, growth, pregnancy, size **>2 cm** or symptoms. An acquired AVF may close within months, so observe it.
- A congenital arteriovenous fistula presents with bruit, haematuria and high-output failure and rarely resolves.

8 · COLLECTING SYSTEM ANOMALIES

P1 *Pelvicalyceal anomalies other than PUJ obstruction — calyceal diverticulum (types I/II, stones ≈40%, delayed contrast pooling, percutaneous ablation and fulguration), hydrocalycosis, megacalycosis (non-obstructive, 12–20 calyces, male 6:1, no surgery), infundibulopelvic stenosis (bilateral, >⅓ renal failure), extrarenal and bifid pelvis (≈10%)*

Not in the paper answer:

- Extrarenal pelvis: confirm with diuretic renography before calling it obstruction.
- Infundibulopelvic stenosis: monitor renal function yearly; if it falls, give an ACE inhibitor and restrict protein.

> **KEY POINT**
> Collecting-system anomalies split on one question: is there obstruction? Dilatation without obstruction (bifid, extrarenal pelvis, megacalycosis) needs nothing; stasis (diverticulum, hydrocalycosis) is treated for symptoms; infundibulopelvic stenosis threatens function through bilateral dysplasia.

9 · LANDMARK TRIALS

Trial	Design	Result	Take-home
RAFT (Renal Anhydramnios Fetal Therapy)	Non-randomised multicentre; isolated BRA, serial amnioinfusion before 26 wk	94% live births; 82% survived ≥14 d with dialysis access; only 35% discharged on peritoneal dialysis; stopped early	Prevents lethal lung hypoplasia but leaves neonatal ESRD — research only

50 Renal Dysgenesis and Cystic Disease of the Kidney

1 · GENETICS IN ONE TABLE

Gene	Renal phenotype
PAX2	Renal hypodysplasia or aplasia
EYA1	Branchio-oto-renal (BOR) syndrome
HNF1B	Renal cysts and diabetes syndrome
PKD1, PKD2 (primary cilium)	Polycystic kidney disease

2 · DYSGENESIS: DYSPLASIA VS HYPOPLASIA

Dysplasia is the leading cause of childhood renal failure.

- **Dysgenesis** (abnormal size, shape or structure) includes dysplasia and hypoplasia.
- **Dysplasia** is a **histological** diagnosis of arrested metanephric differentiation. Its sine qua non is **primitive ducts** with fibromuscular collars; cartilage may also occur. Not all cysts are dysplastic, and not all dysplasia is cystic.
- Causes are primary UB failure (syndromes such as BOR and renal–coloboma) or fetal outflow obstruction. **It is the leading cause of childhood end-stage renal disease (ESRD).** Ultrasound (US) shows a small echogenic kidney with lost corticomedullary differentiation.
- **Hypoplasia** is reduced mass from too few cells; simple hypoplasia has a normal nephron number.

Entity	Defining feature	Course / management
Oligomeganephronia	Few, hypertrophied nephrons; usually bilateral	Polyuria, polydipsia, short stature; **proteinuria first sign**; ESRD in second decade. Fluids, correct salt loss/acidosis, protein 1.5 g/kg/d, ACE inhibitor; living related donor acceptable (not familial)
Ask-Upmark (segmental hypoplasia)	Mostly acquired reflux nephropathy; deep "slit-scars"	Girls ≥10 y, **severe hypertension**; partial/total nephrectomy cures blood pressure (BP); bilateral: medical, dialysis/transplant. Reflux correction does not cure hypertension
Hypodysplasia	With ectopic orifice, megaureter, ureterocele, posterior urethral valves (PUV), prune-belly	**Dysplasia is proportional to the degree of ureteric ectopia**: lateral orifice gives reflux-associated, medial/ureterocele gives obstruction-associated dysplasia

> **KEY POINT**
>
> The ureteric orifice predicts the kidney: a bud from the wrong place on the wolffian duct meets the wrong blastema, so ectopia, obstruction and dysplasia travel together.

3 · CYSTIC DISEASE: CONCEPTS AND CLASSIFICATION

Sort first by inheritance, then by the extrarenal clue.

- Cysts arise from tubules; the **primary cilium** is central.

- A **multicystic** kidney (MCDK) is dysplastic with no normal nephrons. A **polycystic** kidney starts normal, and its nephrons are progressively replaced by cysts.

Inheritable	Non-heritable
Autosomal recessive polycystic kidney disease (ARPKD)	Multicystic dysplastic kidney (MCDK)
Autosomal dominant PKD (ADPKD)	Benign multilocular cyst (cystic nephroma)
Juvenile nephronophthisis (recessive) / medullary cystic disease (dominant)	Simple cysts
Congenital nephrosis; familial hypoplastic glomerulocystic disease	Medullary sponge kidney; sporadic glomerulocystic disease
Syndromes: tuberous sclerosis, von Hippel–Lindau (VHL)	Acquired renal cystic disease; calyceal diverticulum

The extrarenal clue decides: hepatic fibrosis means ARPKD; berry aneurysm, ADPKD; retinitis pigmentosa, nephronophthisis; angiofibroma, tuberous sclerosis complex (TSC); haemangioblastoma, VHL.

4 · ARPKD

Huge echogenic kidneys and an always-fibrotic liver.

P3 *Autosomal recessive polycystic kidney disease. — PKHD1 chr 6, fibrocystin; always congenital hepatic fibrosis; oligohydramnios, pulmonary hypoplasia, 30–50% neonatal death; homogeneously hyperechoic kidneys; scan parents; supportive care; ARegPKD early bilateral nephrectomy warning*

Not in the paper answer:

- In later-onset disease, portal hypertension dominates.
- Nephrectomy costs nephron mass and brings dialysis earlier, so delay it where possible.

5 · ADPKD — PAEDIATRIC FOCUS

In children the key finding is hypertension with normal function.

P2 *ADPKD (Adult polycystic kidney disease) — PKD1 (16, 85%, ESRD 50s) vs PKD2 (4, 15%, ESRD 70s); pain, haematuria ≈50%, stones 20–30%, hypertension ≈50% at 20–35 y; liver cysts, aneurysm 10–30%; Ravine criteria; BP <130/80 with ACEi/ARB, tolvaptan (TEMPO, REPRISE), lipophilic antibiotics, unroofing*

Not in the paper answer:

- **Contiguous gene syndrome**: deletion of PKD1 and the adjacent TSC2 gives TSC with severe early polycystic kidneys.
- A negative US in a young person does **not** exclude PKD2, so use genetic testing after counselling when certainty matters.
- Screen for aneurysm by MR angiography only for family history or high risk. Observe <7 mm; coil or clip if >7–10 mm or symptomatic.

> **KEY POINT**
> ADPKD is a volume disease: cyst growth stretches vessels (hypertension), compresses nephrons and hides stones and tumours. Every therapy is judged by whether it slows total kidney volume.

6 · JUVENILE NEPHRONOPHTHISIS (NPH) / MEDULLARY CYSTIC KIDNEY DISEASE (MCKD)

A salt-wasting cause of renal failure that ultrasound often misses.

- These ciliopathies give tubular atrophy, corticomedullary cysts and interstitial fibrosis in small-to-normal kidneys. **Polyuria and polydipsia occur in >80%**, with **salt wasting**. ESRD follows 5–10 y after presentation.

	Juvenile NPH	MCKD
Inheritance	Autosomal recessive	Autosomal dominant
ESRD	Mean 13 y; **10–20% of childhood renal failure**	Third–fourth decade
Hypertension	Absent	Present
Extrarenal	20%: **retinitis pigmentosa (Senior-Løken)**, Joubert, Cogan apraxia, hepatic fibrosis, Bardet–Biedl	None; type 2 — hyperuricaemia, gout in teenage males

- CT or MRI shows the cysts better than US. **Replace sodium, and avoid salt restriction and diuretics**. Transplant works, with no recurrence.

7 · SYNDROMES WITH RENAL CYSTS

Tuberous sclerosis complex (TSC)

- It is autosomal dominant: **TSC1 (chr 9), TSC2 (chr 16)**. Diagnosis needs 2 major features, or 1 major and 2 minor (angiomyolipoma is major).
- Kidney: cysts and **angiomyolipoma (AML) in 40–80%**, bilateral and multiple. AMLs never metastasise; the risk is haemorrhage. RCC occurs in ≈2%; suspect it in a growing fat-free lesion.
- **MRI is preferred**, because US misses 25–30% of fat-poor AMLs.
- **If an AML is >3 cm and growing, give everolimus (a mammalian target of rapamycin, mTOR, inhibitor) first line**. For an acute bleed, embolise and then give corticosteroids. Kidney-sparing surgery is second line; avoid nephrectomy. Image every 1–3 y, check BP, proteinuria and GFR yearly, and treat hypertension with an ACE inhibitor or ARB.

Von Hippel–Lindau disease

P2 *VHL gene and von Hippel–Lindau disease — AD, 3p25, pVHL degrades HIF, which drives VEGF; lesion frequencies (cysts 76%, clear cell RCC, phaeo, haemangioblastoma, epididymal cystadenoma); diagnostic criteria; VHL Alliance 2020 surveillance (abdominal MRI 2-yearly from 15); nephron-sparing surgery, recurrence 75–85%, DSS 81–94%; partial adrenalectomy*

Not in the paper answer:

- Partial adrenalectomy for a small phaeochromocytoma avoids lifelong steroids.

8 · MULTICYSTIC DYSPLASTIC KIDNEY (MCDK)

The MCDK is harmless; watch the opposite kidney.

- It has non-communicating cysts, **no normal parenchyma, no function** and an atretic ureter ("bunch of grapes").
- It is the **commonest renal cystic disease of the newborn and the second commonest neonatal abdominal mass**: 1 in 1000–4000. Bilateral disease is lethal.
- **The contralateral kidney** hypertrophies and has **vesicoureteral reflux (VUR) in 18–43% and pelviureteric junction (PUJ) obstruction in 3–12%**.
- About 40% involute (later imaged as "agenesis"). Hypertension and Wilms tumour are not increased.

US feature	MCDK	Severe PUJ obstruction
Cysts	Haphazard, varied, non-communicating	Calyces around central pelvis, communicating
Central large cyst	Absent	Present (pelvis)
Isotope uptake	None	Usually some

ALGORITHM

1. For a prenatally detected MCDK, do US on day 2–3 and at 1 month to assess the contralateral kidney and bladder.
2. If the diagnosis is in doubt, do a DMSA or MAG3 scan.
3. If there is a urinary tract infection (UTI) or an abnormal contralateral kidney or bladder, then do a voiding cystourethrogram (VCUG); otherwise do not.
4. Hand over to primary care to check BP, mass and UTI.

- **Nephrectomy** is only for mass effect, hypertension, cyst rupture or **increasing solid tissue**. Routine US surveillance has no proven benefit.

KEY POINT

MCDK removes itself; management looks to the solitary contralateral kidney and its reflux or obstruction.

9 · BENIGN MULTILOCULAR CYST (CYSTIC NEPHROMA)

Imaging cannot exclude cystic Wilms tumour, so excise.

- **It is bimodal: boys <4 y (M 2:1, flank mass) and women >30 y (F 8:1, haematuria from herniation into the pelvis)**.
- **Excision is both diagnosis and treatment** (nephrectomy in children; partial if localised).

10 · SIMPLE CYSTS IN CHILDREN AND BOSNIAK

- **US criteria (all four)**: anechoic, thin smooth wall, posterior enhancement, round or oval. On CT a simple cyst is −10 to +20 HU without enhancement. A hyperdense cyst is benign if non-enhancing, homogeneous and ≤3 cm.
- **Children**: if a simple cyst is <15 mm and asymptomatic, repeat US at 2 y and discharge if stable. Continue follow-up if it is ≥15 mm or complex, or with a family history or genitourinary anomalies.

Bosniak	Features	Management
I	Simple	None
II	Fine septa, minimal calcification, hyperdense ≤3 cm	None
IIF	More septa/calcification, no measurable enhancement	Follow-up (≈11% progress)
III	Thick/irregular enhancing walls or septa	Excision (paediatric III–IV: >90% intermediate/malignant)
IV	Enhancing solid components	Excision

Treat a symptomatic benign cyst by aspiration with sclerosis or laparoscopic unroofing.

11 · MEDULLARY SPONGE KIDNEY, GLOMERULOCYSTIC AND ACQUIRED CYSTIC DISEASE

P2 *Medullary sponge kidney — ectatic collecting ducts, urographic triad ("brush bristles", "bouquet of flowers"), presents >20 y, colic 50–60%; hypercalciuria ⅓–½, incomplete dRTA 30–40%; potassium citrate first, thiazide, phosphate (not with urease infection)*

- **Sporadic glomerulocystic disease** is a diagnosis of exclusion.

P2 *Acquired renal cystic disease — ≥3 cysts/kidney in ESRD; uraemia not dialysis; 10% pre-dialysis, rising to >90% at 10 y.*

Small kidneys; haemorrhage, adenomas, RCC (aggressive in native kidneys after transplant); bleeding: embolisation/nephrectomy; tumour >3 cm excise, <3 cm nephrectomy or yearly CT.

Not in the paper answer:

- Acquired renal cystic disease affects 20–45% of children on dialysis, and a retroperitoneal bleed calls for nephrectomy because an occult RCC is likely.
- **Parapelvic cysts** mimic hydronephrosis on US; delayed contrast CT shows an undilated system, and management is conservative.

12 · LANDMARK TRIALS

Trial	Design	Result	Take-home
HALT-PKD	Randomised: lisinopril and telmisartan vs lisinopril; rigorous vs standard BP	Dual blockade no benefit; low BP slowed total kidney volume (TKV) growth, lowered left ventricular mass and albuminuria, GFR unchanged	Single renin–angiotensin blocker; tight BP early
TEMPO / REPRISE	Randomised, tolvaptan vs placebo	Slower TKV growth and GFR decline	First disease-modifying drug; aquaresis, hepatotoxicity
ARegPKD registry	Observational ARPKD cohort	Bilateral nephrectomy <3 months gave the worst neurological outcome	Avoid early bilateral nephrectomy

51 Pathophysiology of Urinary Tract Obstruction

1 · DEFINITION AND CLASSIFICATION

Function, not dilation, defines obstruction.

- **Obstruction** is any impediment to flow, calyces to meatus, that injures the kidney or lower tract: **acute or chronic, partial or complete, intrinsic or extrinsic, unilateral or bilateral**.
- It affects ≈ **1 in 500** live births; high-grade obstruction carries a lifetime end-stage renal disease (ESRD) risk of up to **30%**.

	Lower urinary tract obstruction (LUTO)	Upper tract
Commonest	Posterior urethral valves (PUV) and urethral atresia ≈ 50%	Ureteropelvic junction obstruction (UPJO), 1 in 1500; M:F 2:1
Others	Anterior urethral valves, megalourethra, prolapsing ureterocele, cloaca, megacystis–microcolon–intestinal hypoperistalsis, trisomy 18/21	Ureterovesical junction obstruction (UVJO), ectopic ureter, ureterocele, polyp, stones, crossing vessel
Outcome driver	ESRD and pulmonary hypoplasia	Unilateral function loss; contralateral kidney buffers

Presentation is prenatal hydronephrosis (mostly at 18–20 wk), or later infection, mass, pain or respiratory failure.

> **KEY POINT**
>
> Trap: **detrusor hypertrophy** causes UVJ obstruction by narrowing the intramural ureter. Neurogenic bladder obstructs without an anatomical lesion.

2 · HAEMODYNAMIC CHANGES

Rising pressure lowers GFR and renal blood flow (RBF), causing ischaemia. Injury begins within **2 h** of complete obstruction, worse if bilateral or solitary.

† Phase (complete)	Unilateral	Bilateral / solitary
I (0–1.5 h)	RBF rises, ureteric pressure rises (afferent dilatation: PGE2, nitric oxide)	Modest RBF rise

† Phase (complete)	Unilateral	Bilateral / solitary
II (1.5–5 h)	RBF falls, pressure still high (efferent constriction)	RBF falls, pressure rises
III (> 5 h)	RBF and pressure fall (afferent constriction: angiotensin II, thromboxane)	Pressure stays high (retained solute, ANP), the basis of post-obstructive diuresis

- **Hypertension** is renin driven: bilateral ≈ **77%**, unilateral ≈ **20%**.

3 · TUBULAR FUNCTION

- **Aquaporins and sodium transporters are downregulated**, causing a concentrating defect, salt wasting, polyuria and **distal renal tubular acidosis**.
- Potassium and other solutes are **wasted early and retained late**. FENa is < 1% in acute obstruction and > 1% with tubular injury.

4 · PATHOLOGY AND FIBROSIS

- Fibrosis is the final common path, driven by stretch and **TGF-β/Smad**, amplified by angiotensin II.
- **Bladder** hypertrophies, compensates, then decompensates with poor compliance. Storage pressure > **40 cm H_2O** injures the kidneys.

5 · RECOVERY AND COMPENSATORY GROWTH

- Relief can reverse damage, but **incomplete recovery is common**.
- Compensatory growth is **glomerulomegaly, not new nephrons**.

6 · POST-OBSTRUCTIVE DIURESIS (POD)

P1 *Post-obstructive diuresis — unilateral ≈ **2%**, bilateral ≈ **50%**; aquaporin/transporter loss, medullary washout, retained solute; risk after valve ablation (oligohydramnios, preterm, creatinine); monitoring, partial replacement, avoid overcorrection*

Not in the paper answer:

- Preterm infants have low aquaporin-2, hence their POD risk.

ALGORITHM

1. Monitor intake, output, and serum and urine electrolytes.
2. Replace losses and correct electrolytes.
3. **Do not over-replace**: it perpetuates POD.
4. Expect resolution within **24 h**; it seldom lasts > 48 h.

7 · FETUS AND CHILD

P3 *Antenatal hydronephrosis (UTD) and fetal uropathy (LUTO) — UTD APD thresholds (**≥ 7 mm** at 16–27 wk, **≥ 10 mm** ≥ 28 wk), serial fetal urine (Na < 100, Cl < 90, osmolality < 200, β2-microglobulin < 6)*

Fetal intervention: shunt/fetal cystoscopy, amnioinfusion

Not in the paper answer:

- Second-trimester oligo-or anhydramnios carries **60–80%** mortality.
- In the Whitaker test, obstruction is a pelvic pressure > **12 cm H_2O** or a continuous rise.
- After birth, use ultrasound first, MAG3 for drainage, DMSA for scars and VCUG for outlet and reflux.

8 · TREATMENT PRINCIPLES

Drain to buy function, then reconstruct.

Option	Indication / key point
Clean intermittent catheterisation (CIC)	Neurogenic bladder; reduces UTI, stones, retention
Overnight catheter drainage	Valve bladder with nocturnal polyuria and progressive CKD
Vesicostomy	Infant with complex LUTO, unstable or unable to do CIC
Percutaneous nephrostomy	Upper tract; needs some hydronephrosis
Stent / cutaneous ureterostomy	Temporising; stent excellent for extrinsic obstruction
Pyeloplasty / reimplantation	Definitive; anastomose anterior to aberrant vessel; non-refluxing reimplant in children

- Use NSAIDs only if function is preserved. ESRD needs dialysis or transplant (Ch 61).

52 Ectopic Ureter, Ureterocele and Ureteral Anomalies

1 · DEFINITIONS AND CLASSIFICATION

- Ectopic ureter and ureterocele are one defect of **ureterotrigonal connection** on a continuum. The appearance changes the technique, not the strategy.
- An **ectopic ureter** is a single or duplex ureter that does not enter the trigone but enters **at or distal to the bladder neck**; merely opening near it is not ectopic.
- A **ureterocele** is **cystic dilation of the distal ureter** within the bladder or spanning bladder neck and urethra. It is never entirely urethral and never attached to wolffian structures.
- In a **duplex system, both lesions always involve the upper pole** ureter.

	Ectopic ureter	Ureterocele
Distal ureter	Dilated or not, opens ectopically	Cystic dilation
Position	At/distal to bladder neck; perineum, vagina, rectum (F) or wolffian ducts (M)	Bladder or spanning bladder neck and urethra
Wolffian attachment	Yes (male)	Never
Incontinence (untreated)	Yes, in females	No

Ureterocele classification

P3 *Ureterocele — Glassberg intravesical vs ectopic (most useful, predicts TUI outcome); Stephens incl. cecoureterocele; Churchill functional*

Not in the paper answer:

- A **cecoureterocele** tunnels submucosally beyond the bladder neck into the urethra; it is often missed before surgery and risks incontinence.
- **Ureterocele disproportion** is a typical ureterocele with a small, dysplastic, non-dilated upper pole, so the kidney looks normal.

2 · EMBRYOLOGY ESSENTIALS

Bud position explains upper pole obstruction and lower pole reflux.

- In a duplex, the upper pole bud arises **later and more cephalad** on the wolffian duct, so its orifice is carried **caudally** and becomes ectopic.
- **Weigert–Meyer rule**: **the upper pole orifice lies caudal (and medial) to the lower pole orifice**. So the upper pole gets ectopia or ureterocele (obstruction), and the lower pole gets a short

tunnel and vesicoureteral reflux (VUR).

- Classically, a ureterocele results from failure of the **Chwalle membrane** to rupture. Renal dysplasia lies **downstream** of the connection defect.
- An ectopic ureter reaches the vagina, cervix or uterus only through the **Gartner duct**, never directly into müllerian structures.

3 · ECTOPIC URETER — SITES AND SEX ASYMMETRY

P3 *Embryology and management of ectopic ureter. — female vs male sites, continuous dribbling vs never incontinent, epididymitis, Zinner syndrome, Gartner duct*

Not in the paper answer:

- A **single-system ectopic ureter** may mimic an "absent kidney" on ultrasound; use contrast CT or a renal function study.
- **Bilateral single-system ectopia** (rare) gives a hypoplastic bladder, poor bladder neck and bilateral dysplasia.

> **KEY POINT**
> Why does only the girl leak? The female ureter may open distal to the external sphincter and leaks continuously; the male ureter, confined to wolffian derivatives above it, presents as infection or obstruction.

- Male sites are the bladder neck, prostatic urethra, vas, seminal vesicle and ejaculatory duct.

4 · CLINICAL PRESENTATION

Prenatal

- Most are found as prenatal hydronephrosis. A fetal upper pole "cyst" is upper pole hydronephrosis until proven otherwise.
- **The diagnosis is made at the bladder**, because the upper tracts look identical; a large ureterocele can mask the bladder.
- Dilation of the ipsilateral lower pole or contralateral kidney suggests reflux. Dilation of the whole kidney and ureter suggests a single system.

Incidental and infection

- A "presumed ovarian cyst" may be a dilated ureter, so include it in the differential of a cystic pelvic mass.
- Do ultrasound in every child with urosepsis. If sepsis does not respond, do an **urgent transurethral incision (TUI)** of a ureterocele, or drain an ectopic ureter (usually end ureterostomy).
- An infected ectopic ureter gives low-grade spiking fever, and a **negative culture does not exclude infection**.
- **Any young male with bacterial epididymitis needs upper tract ultrasound**; it is often recurrent before diagnosis.

Incontinence

- Suspect it in a toilet-trained girl with **continuous low-volume dampness day and night** and normal voiding. Untreated ureteroceles do not cause incontinence.
- The key question is: **can she be dry for 30–60 minutes?** With an ectopic ureter she cannot.
- The leaking moiety may be non-dilated and missed on imaging, so suspicion rests on the history. Exclude voiding postponement and constipation.

Ureterocele prolapse

- It appears as a **smooth, congested, mucosa-covered interlabial mass**, often with voiding difficulty. A lobulated grape-like mass suggests **sarcoma botryoides** instead.

- Ultrasound confirms it; VCUG shows the prolapse, lower pole reflux and the "drooping lily".

Late presentation

- Teenagers and adults present with infection or pain; an adult single-system ureterocele often contains a **stone**. Upper pole heminephrectomy is the mainstay.

5 · EVALUATION

- Examine the perineum of every continuously wet girl. The ectopic orifice lies **between the urethral meatus and the vagina**.

Imaging

Test	Answers	Note
Ultrasound	Anatomy; inferred function; bladder	First always; cannot separate ectopic ureter from ureterocele above bladder
Voiding cystourethrogram (VCUG)	Reflux (lower pole, contralateral, into ureterocele); bladder base, urethra	Optional if obstruction alone dictates incision; repeat after incision
Dimercaptosuccinic acid (DMSA)	Differential function — gold standard	Only "some vs none"
Mercaptoacetyltriglycine (MAG3) renogram	Function and drainage	Observation candidates; after 2 months
MR urography	Most detailed anatomy and function	Complex anatomy, non-dilated upper pole; needs sedation
Intravenous pyelogram (IVP)	Anatomy and qualitative function	Non-dilated ectopic ureter with incontinence

- A **ureterocele** is a thin-walled cyst within the bladder. A **pseudoureterocele** is a dilated ectopic ureter indenting the bladder; its wall is **bilaminar and thicker, and its lumen lies outside the bladder**.
- A full bladder can efface a ureterocele so it mimics an ectopic ureter, so scan through filling and voiding.
- On VCUG, the **"drooping lily"** is the lower pole pushed down and out by a dilated upper pole.
- † A single-system intravesical ureterocele gives a **"cobra head" (spring-onion)** sign on urography.
- **Preoperative function does not predict function after decompression**.

VCUG terms

Term	Meaning	Significance
Effacement	Flattens with filling	Mimics ectopic ureter
Prolapse	Moves into urethra	Outlet obstruction; predicts secondary surgery
Eversion	Diverticulum-like protrusion of bladder base	**Weak trigonal floor, so open repair likely**

Endoscopy

- **Choose the incision site** with little bladder filling, when the lowest part of the ureterocele is seen best.
- Detect any urethral extension (cecoureterocele) so the incision leaves no obstructing segment.
- A patulous bladder neck or cecoureterocele carries a later incontinence risk; consider bladder neck tightening.

6 · MANAGEMENT PRINCIPLES

- Goals: **preserve renal function; eliminate infection, obstruction and reflux; maintain continence; minimise total morbidity**.

- Start antibiotic prophylaxis early in significant hydronephrosis, even without reflux.
- Relieving obstruction may **create** reflux, or may let lower pole reflux resolve. This is the central trade-off, and decisions are empirical.
- **Fetal intervention**: none for ectopic ureter; for ureterocele only if it obstructs all units (experimental).

Neonatal algorithm

ALGORITHM

1. After prenatal hydronephrosis, do postnatal ultrasound.
2. If there is significant hydroureteronephrosis, start prophylaxis.
3. Do VCUG (optional if it will not change the decision to decompress), then a renal scan after 2 months.
4. If the ureterocele is small, with minimal hydronephrosis and no obstruction, observe.
5. If obstruction is significant and parenchyma is at risk, do TUI (or cutaneous distal ureterostomy if endoscopy is impossible).
6. If sepsis does not respond, do TUI (ureterocele) or nephrostomy or end ureterostomy (ectopic ureter); elective drainage is still needed later.

- **Observation** suits selected ureteroceles: no obstruction of the lower pole or contralateral kidney and lower pole reflux ≤ grade III, a non-functioning upper pole, or no obstruction on renogram. Counsel about unpredictable acute presentation.

7 · OPERATIVE OPTIONS

Transurethral incision (TUI) of ureterocele

- TUI relieves obstruction in **78–97%**. It is the choice in **sepsis** and in the **infant**.
- Make a **low transverse full-thickness incision as distal and close to the bladder floor as possible** to reduce reflux. For an ectopic ureterocele, extend it into the urethral part so no obstructing flap remains.
- The **"watering-can"** technique (multiple holmium punctures) decompresses equally with less de novo reflux.
- It is a day-case procedure. Do ultrasound and VCUG at **4–6 weeks**.
- New reflux occurs in **0–50%**. **Intravesical** ureteroceles decompress without reflux in 70–80%; **extravesical** ones need secondary surgery in up to 70%. Lower pole reflux (risk ratio 1.74) and an extravesical site (2.78) predict reoperation.

> **KEY POINT**
>
> Why TUI first even if half need more surgery? It is outpatient, may be definitive, defers the second operation to an older child, and decompression often avoids tapering at reimplant. It is less appealing in an older child with a massive non-functioning upper pole, who needs definitive surgery at diagnosis.

Upper pole partial nephrectomy (heminephrectomy)

- It is for a non-functioning upper pole, by open, laparoscopic or robotic approach. The main concern is **not to damage the viable lower pole**.
- Use the upper pole ureter as a handle, passing it **behind the renal vessels**, ligate the upper pole vessels and resect at the demarcation.
- **Dissect on the wall of the upper pole ureter**, and leave its back wall on the lower pole ureter near the bladder, because they share a blood supply. A small distal stump is left; a large or refluxing one is excised low, sparing the bladder neck and vas.
- Outcomes: for ectopic ureter it is usually curative. For ureterocele, new reflux appears in **15–50%** and secondary surgery is needed in **40–50%**.

Ureteroureterostomy / pyeloureterostomy

- The upper pole is joined end-to-side into the lower pole system, for functioning and increasingly non-functioning upper poles; feared hypertension has not been substantiated.
- For a **distal** repair, **stent the lower pole ureter cystoscopically first**. A **proximal** repair is preferred for a dilated upper pole or size mismatch. Robotic repair matches open results, and size mismatch is no barrier.

Lower tract reconstruction

- **Ureterocele excision with common sheath reimplantation** relieves obstruction and reflux at once, and is the fallback for persistent reflux. It risks the bladder neck, sphincter and vagina.
- Mobilise **both ureters as a unit**, excise the ureterocele, **plicate the detrusor defect** and reimplant submucosally.
- A cecoureterocele remnant can act as an obstructing **urethral windsock** flap-valve.
- Reflux persists in **5–10%**, more often after tapering.
- **Total reconstruction** suits an **older child with a massive ureterocele, non-functioning upper pole and significant lower pole reflux**.

End ureterostomy for ectopic ureter

- It is for a neonate with sepsis or massive dilation. **Do not trim the redundant ureter**, because its length is needed for reimplant. Do DMSA after a few months, then reconstruct or remove.

8 · DECISION SUMMARY

Upper pole function and lower pole reflux pick the operation.

Ectopic ureter (duplex)

ALGORITHM

1. If the upper pole is worth keeping and the lower pole refluxes, do a common sheath reimplant, or a lower pole reimplant with distal ureteroureterostomy.
2. If the upper pole is worth keeping and there is no lower pole reflux, do a proximal or distal ureteroureterostomy.
3. If function is unclear, or there is massive dilation or sepsis, do a temporary end ureterostomy, then DMSA, then reconstruct or remove.
4. If removal is chosen, do upper pole nephrectomy with ureterectomy.

Ureterocele (Table 52.1)

Procedure	Ideal indication	Advantage	Limitation
TUI	Infant; large ureterocele with VUR	Outpatient; effective; occasionally definitive	Reflux into ureterocele moiety needs bladder surgery
Upper pole nephrectomy	Older; large non-functioning upper pole; no VUR	Removes pathology; avoids bladder	Lower pole risk; may still need bladder surgery
Ureteroureterostomy / pyelostomy	Older; functioning upper pole; no VUR	Drains with little risk	Ureterocele left; may reflux
Excision and common sheath reimplant	VUR; functioning upper pole without major dilation	Removes obstruction and reflux; no renal risk	Complex; vagina/bladder neck risk; tapering

KEY POINT

VUR is the pivot: without VUR choose an upper tract solution; with VUR, a bladder-level solution, except in the infant, where TUI defers bladder surgery. Upper pole function then picks nephrectomy or ureteroureterostomy.

9 · COMPLICATIONS

- **Loss of lower pole function** after heminephrectomy is the most significant; prevent it by preserving the shared blood supply and using gentle traction.
- **Urinoma** occurs in up to 20% after laparoscopic or robotic cases; observe it if asymptomatic.
- **De novo reflux** after TUI or heminephrectomy is observed if there is no infection; otherwise do excision and reimplantation.
- **Voiding dysfunction** comes from poor trigonal support, which causes obstruction, poor emptying and recurrent VUR. Assess with VCUG **including lateral voiding views**, and treat by reconstruction, CIC or bulking.

10 · OTHER URETERAL ANOMALIES

Duplication

- It occurs in ~**1 in 125** (0.8%), F:M 1.6:1, and is mostly unilateral. **Look for contralateral duplication**: an occult second ectopic ureter explains persistent wetting.
- A **bifid ureter** is a partial duplication joining below the ureteropelvic junction (UPJ).
- **The lower pole gets single-system disease** (UPJ obstruction, VUR); **the upper pole gets ureteral-formation disease** (ectopia, ureterocele).
- Lower pole VUR is often high grade and resolves slowly; treat by common sheath reimplant, endoscopic injection or low ureteroureterostomy.

Triplication

- Smith types: 1, three separate ureters (**35%**, commonest); 2, incomplete with two orifices (21%); 3, trifid with one orifice (31%); 4, two ureters with three orifices (13%).

Valves, diverticulum, polyp

- † **Ureteral valves** are transverse mucosal folds with smooth muscle and proximal dilation; excise the segment. A ureteral **diverticulum** is usually a **blind-ending bifid ureter**.
- A **fibroepithelial polyp** is commonest at the **UPJ**, **89% male, 78% left**. At the UPJ, do pyeloplasty; elsewhere, **remove it ureteroscopically**.

Vascular position anomalies

P1 P2 *IVC anomalous development and retrocaval ureter; Retroileal Ureter — persistent subcardinal vein, right only, type I fishhook vs II, CT urography, transpose anterior; retroiliac at L5–S1 with mesonephric ectopia*

Not in the paper answer:

- A retrocaval ureter is always right-sided and mimics UPJ obstruction, so a simple pyeloplasty fails.

53 Surgery of the Ureter in Children

1 · PELVIURETERIC JUNCTION (PUJ) OBSTRUCTION — DEFINITION AND AETIOLOGY

P1 *Describe Pathogenesis of Congenital Pelvi Ureteric Junction Obstruction its diagnosis and Management. — Koff definition; intrinsic (infant) vs crossing vessel (older, intermittent pain) vs secondary; bilateral 10–40%; associated anomalies up to 50%; SFU grades; MAG3 T½ cut-offs; one-third*

need surgery

Not in the paper answer:

- Paediatric hydronephrosis is a pelvic anteroposterior diameter (APD) of **12 mm**, but hydronephrosis is not obstruction.

- With a crossing vessel, **dismembering and transposing the ureter anterior to the vessel prevents recurrence**. For secondary PUJ obstruction from high-grade reflux, avoid reimplant and pyeloplasty at one sitting; stage them.
- For **lower-moiety PUJ obstruction in a duplex**, do ureteropyelostomy or pyeloplasty; if the moiety has ≤ **5–10%** function, do partial nephrectomy.

2 · DIAGNOSIS AND INDICATIONS

Decide on function trend and dilatation, not on washout alone.

- Most are found on antenatal ultrasound (US) at ≈ 20 weeks.
- Neonatal renogram washout has false positives up to **44%**, so decide on the trend in differential renal function (DRF) and on dilatation, not on half-time (T½) alone.
- **Pyeloplasty indications**: increasing APD; low or falling DRF; breakthrough infection on prophylaxis; symptoms (pain; failure to thrive in infants; haematuria; rarely hypertension).

3 · ANDERSON–HYNES DISMEMBERED PYELOPLASTY

It is the **gold standard** for any access, with success ≈ **95%**. It preserves a crossing vessel, excises the adynamic segment and allows pelvic reduction.

ALGORITHM

1. Expose the PUJ and free the lower pole, looking for an accessory vessel.
2. Use bipolar diathermy sparingly and place stay sutures.
3. Decompress the pelvis with a 21-G needle, then dismember, with or without pelvic reduction.
4. **Spatulate laterally** into healthy ureter, keeping the stenotic segment as a "no-touch" handle.
5. Bring the apex of the spatulation to the inferior pelvis and make a tension-free, watertight 6-0/5-0 absorbable anastomosis.
6. Excise the stenotic segment last, irrigate out clot and cover with perinephric fat.

- For a long gap, mobilise the whole kidney. A drain is usually unnecessary.

Alternative	Best indication	Point
Foley Y-V-plasty	High insertion, **no vessel**	V base on dependent medial pelvis, apex at PUJ
Spiral flap	Long proximal stricture, dependent PUJ	Length:width ≤ 3:1
Vertical flap	Dependent PUJ, large box-shaped extrarenal pelvis	Flap length must match gap
Vascular hitch (Hellström)	Anterior lower-pole vessel, **no intrinsic stenosis**: moderate hydronephrosis, no calyceal dilatation, good cortex and split function; peristalsing PUJ at surgery	Vessel fixed cranially in pelvic wall; no hypertension seen
Ureterocalicostomy	Failed repair, short scarred ureter, dilated lower calyx with thin cortex	Lower-pole nephrectomy exposes calyx; spatulated ureter joined over stent

Use flaps only when a tension-free primary anastomosis is impossible. Non-dismembered repair is inferior.

KEY POINT

Every variant answers two questions: is there a crossing vessel (dismember), and is the gap too long for tension-free anastomosis (flap, mobilisation)?

KEY POINT

One fixed principle, many routes: dismember, excise, spatulate, anastomose without tension. Failure is managed by repeating the principle.

4 · ACCESS

- In **infants**, open surgery is preferred; **posterior lumbotomy** suits only infants and toddlers, while the flank approach suits any age.
- Laparoscopic results equal open, and robotic surgery has a shorter learning curve.

	Transperitoneal	**Retroperitoneal**
For	Easier landmarks; left transmesocolic	No bowel injury; leak extraperitoneal; shorter stay
Against	Bowel risk; urinary ascites causing ileus	Learning curve; small space
Note	Preferred for horseshoe kidney, retrocaval ureter	One randomised trial favoured it; table **not flexed**

- **Endopyelotomy** does poorly long-term under 18 years; its niche is **failed pyeloplasty** without a vessel, with a double-J stent for 6 weeks.

5 · STENTS, OUTCOMES, FAILURE

Stent	**Removal**	**Point**
Internal double-J	Second anaesthetic unless string/magnet	Catheter 24–48 h; string gives more febrile UTI (girls)
Externalised nephroureteric	7–14 days, no anaesthesia	Fewer complications, shorter stay in infants
Stentless	—	Needs flawless technique

- Stent duration of 1–4 weeks does not change outcome.
- For emergency diversion before repair, use a **nephrostomy, not a retrograde stent**, because a stent inflames the PUJ.
- Success is 90–100%; the commonest reason for conversion is bleeding.

P1 *Management of failed pyeloplasty — early failure = oedema/clot: wait with or without nephrostomy; late: endopyelotomy or redo transperitoneal pyeloplasty (best); ureterocalicostomy for short scarred*

ureter

6 · PRIMARY MEGAURETER

Most resolve; the challenge is spotting the truly obstructed.

P3 *Primary obst. megaureter. — ≥ 7 mm; four types; adynamic juxtavesical segment maturing craniocaudally; < 10 mm rarely needs surgery; indications incl. DRF < 40% or fall ≥ 5%; temporising options; excise–taper–5:1 reimplant; Starr/Kalicinski/Hendren; success > 90%*

Not in the paper answer:

- In infancy the small bladder cannot take a safe tunnel for a massive ureter, so temporise with a **distal loop cutaneous ureterostomy** or a **refluxing reimplant**; balloon dilatation has weak evidence.
- Follow into adulthood, since some worsen at puberty.

> **KEY POINT**
>
> Why not rely on the renogram? Neonatal washout depends on immature function, hydration and a huge ureter's capacity, so serial DRF, diameter and infection history drive surgery. Megaureter is the PUJ problem moved to the bottom of the ureter: an immature adynamic segment that usually matures.

ALGORITHM

1. Excise the stenotic distal segment and straighten the ureter.
2. Taper if needed.
3. Reimplant non-refluxing at a **5:1** tunnel length:diameter.
4. Stent with a feeding tube for 7 days or a double-J for 4 weeks.

- Plication and folding preserve the blood supply; Hendren excisional tapering risks ischaemic stricture.
- Treat postoperative obstruction with a stent first; an ischaemic stricture needs redo.

7 · SURGERY FOR VESICOURETERAL REFLUX

KEY POINT

Principles: exclude or treat bladder causes first; handle ureter and bladder gently; mobilise without tension; keep a tunnel length:diameter of **5:1** (Paquin); ensure a straight course with the bladder empty and full.

- Techniques are intravesical or extravesical, with the tunnel suprahiatal or infrahiatal.

Technique	Approach / tunnel	Strength	Weakness
Politano–Leadbetter	Intravesical; suprahiatal neo-hiatus, vertical tunnel	Long tunnel for high grade	Blind neo-hiatus (bowel, vas, vagina); kink if too lateral/anterior
Glenn–Anderson	Intravesical; infrahiatal toward bladder neck	Least kinking	Short tunnel
Cohen	Intravesical; cross-trigonal	Commonest; success **98%**; thick bladder; with bladder-neck reconstruction	Difficult later retrograde catheterisation

Technique	Approach / tunnel	Strength	Weakness
Lich–Gregoir	Extravesical detrusorotomy ≈ 5 cm	No haematuria/spasm; short catheter	Transient voiding dysfunction/retention up to **10%**, esp. bilateral
Robotic extravesical (RALUR-EV)	Y-shaped detrusorotomy 4–5 cm	Less pain, spasm, stay	Early series: leak, obstruction, ureteric injury

Cohen — key steps

ALGORITHM

1. Open the bladder vertically and put a feeding tube in the orifice as a handle.
2. Circumscribe the orifice and free the ureter, preserving its distal blood supply, until it reaches the contralateral wall without tension.
3. Make a submucosal tunnel across the trigone, with the new orifice above the contralateral one.
4. Suture mucosa to mucosa with 6-0, with a detrusor hitch at 6 o'clock.

Lich–Gregoir — key steps

ALGORITHM

1. With the bladder one-third full, divide the obliterated umbilical artery.
2. Free the ureter **hugging it**, only as far as the UVJ.
3. Distend the bladder and mark a 5-cm tunnel; incise the detrusor down to mucosa and raise wide flaps.
4. Lay the ureter in and close the detrusor caudocranially, catching adventitia at the apex.
5. Leave a Foley catheter for 12–48 h.

KEY POINT

Why voiding dysfunction? The pelvic plexus lies ≈ 1.5–2 cm dorsomedial to the UVJ, so stay on the ureter and stop at the UVJ.

- In the American Urological Association (AUA) meta-analysis, open success is **95.9%**; minimally invasive results are converging.
- **Follow-up** is US at 6–12 weeks. A VCUG is needed for high grade, new hydronephrosis, UTI or a dysfunctional bladder.

Complications of reimplantation

	Early	Long-term
Reflux	Ipsilateral low grade usually resolves by 1 year. Contralateral ≈ 1 in 10 after unilateral repair, mostly resolves, so observe; intervene for pyelonephritis. Routine bilateral repair not justified	Short tunnel/untapered ureter, or **unrecognised secondary reflux** (neurogenic/valve bladder): treat the bladder first (anticholinergic and clean intermittent catheterisation)
Obstruction	Oedema, clot, haematoma, kink within 2 weeks: double-J or nephrostomy; most resolve	Ischaemia, angulation (hiatus too lateral/anterior), tight tunnel, stenosis: balloon and stent, then re-reimplant

- Contralateral reflux risk rises with age < 6 years, capacity < 50% of expected and a refluxing duplex.
- At **reoperation**, excise ischaemic ureter back to healthy tissue; for a short ureter use a **psoas hitch**, Boari flap or transureteroureterostomy.

KEY POINT

Early complications are tissue reactions, so drain and wait. Late ones are design errors (blood supply, hiatus position, tunnel ratio) or an untreated bladder.

Endoscopic injection

P3 *Endoscopic correction of VUR (STING procedure): indications and complications — Dx/HA standard, ≈ 90% low grade; mound within 0.1–0.2 mL, volcano; never cauterise; STING vs HIT vs double HIT; late obstruction (Vantris); Swedish trial 20% recurrence at 2 years*

Not in the paper answer:

- After failure, repeat the injection once or twice, then reimplant; the encapsulated implant does not make reimplantation harder.

8 · OTHER URETERIC LESIONS

- † **Retrocaval ureter** is right-sided with a fish-hook shape; re-anastomose it in front of the cava.
- A **congenital mid-ureteric stricture** is often misread as megaureter; excise it and do a ureteroureterostomy.
- A **fibroepithelial polyp** (upper third, male, left) causes pain and haematuria; remove it endoscopically.
- **Retroperitoneal fibrosis** is treated with steroids, then azathioprine, then a stent, and finally ureterolysis.

56 Hypospadias

1 · DEFINITION, DIAGNOSIS AND EMBRYOLOGY

- The **triad** is an ectopic ventral meatus, ventral curvature and a dorsal hooded foreskin (the last two not always present). Incidence is ≈ **1 in 150–300** male births.

- **Megameatus with intact prepuce (MIP)**, ≈ 5%, hides under a normal foreskin, so retract before newborn circumcision. Avoid circumcision when curvature or ventral skin deficiency is present.

P3 *Embryology of the penis and management of proximal hypospadias (including Koyanagi repair) — SRY, testosterone, 5α-reductase type 2, DHT at 8–12 weeks; endodermal two-zipper vs outdated Glenister ectodermal theory; dorsal prepuce fusion first*

Not in the paper answer:

- The critical window is **weeks 8–16**. Ventral prepuce fusion awaits urethral fusion, so the foreskin is incomplete in **95%**. MCQ

> **KEY POINT**
> Hypospadias is an arrest of the androgen-dependent closing zipper: the meatus stops where closure stopped, the prepuce cannot close over an open urethra, and dysplastic ventral tissue tethers the shaft into curvature.

2 · AETIOLOGY

It is **multifactorial**, but all causes mean too little androgen action in weeks 8–16.

- Placental insufficiency and growth restriction double the risk; assisted reproduction is an independent risk.
- Twin concordance is 9–27%, and chromosomal abnormality is found in up to 30%; about 200 syndromes include it. Endocrine disruptors are unproven in humans.
- It belongs to **testicular dysgenesis syndrome**, and a shorter anogenital distance (AGD) means more severe disease.

3 · CLASSIFICATION AND SEVERITY

- Distal forms are **70–85%**. The meatus is the least reliable marker, so **classify after degloving** by meatus, curvature and plate.

P4 *Hypospadias: present concepts in management, and hypospadias fistula — GMS score table (glans, meatus, shaft; total 3–12, higher means more complications); POST low ratio favours inlay graft over plain TIP*

Not in the paper answer:

- The glans–meatus–shaft (GMS) score (Merriman 2013) is scored in theatre after degloving.

4 · ASSOCIATED ANOMALIES AND DSD WORK-UP

- Hernia or hydrocele occurs in 9–16%, and undescended testis (UDT) in ≈ 7%.
- A **DSD work-up** is needed for UDT with proximal hypospadias, a perineal meatus or atypical genitalia. DSD risk is ≈ **50%** with a non-palpable vs 15% with a palpable testis, and 64% in proximal vs 7% in distal cases. Since 2006 severe hypospadias is classed as 46,XY DSD.
- Do ultrasound for Müllerian remnants, karyotype, testosterone, FSH, LH and AMH; a 46,XY karyotype does not exclude DSD (Ch 59).
- A large **prostatic utricle** causes stasis, UTI and difficult catheterisation. Routine VCUG and renal ultrasound are not needed.

> **KEY POINT**
> Hypospadias with cryptorchidism signals a systemic androgen problem, hence karyotype before surgery.

5 · PRE-OPERATIVE ASSESSMENT

- The goals are a straight erection, a urethra to the tip, good cosmesis and the fewest procedures. Exclude micropenis, and note that a deflected stream may justify surgery even when distal.

- **Timing**: the AAP advises 6–12 months, and the authors **6–18 months**, before genital awareness; younger repair has fewer complications.
- A glans width **<14 mm** means more complications. Give testosterone cypionate **2 mg/kg intramuscularly at 5 and 2 weeks** before surgery if the glans is <15 mm, which gains about 4 mm. Two randomised trials showed fewer complications, but meta-analyses are inconclusive; side effects resolve within 6 months.

6 · INTRA-OPERATIVE PRINCIPLES

- Use a regional block, ketorolac 0.5 mg/kg, antibiotics, and a tourniquet or epinephrine. Cystoscopy is only for severe proximal or redo cases.
- Rapidly absorbed fine sutures beat polydioxanone, whose slow absorption favours fistulae. Close tension-free with non-overlapping lines, and **never use hair-bearing skin**.

7 · CURVATURE

Curvature decides one-stage or staged repair.

It comes from ventral skin tethering, dysplastic dartos, a short urethra and corporal disproportion.

ALGORITHM

1. Deglove to the penoscrotal junction and excise dysplastic dartos.
2. Make an **artificial erection**.
3. If straight, then proceed to urethroplasty.
4. If residual curvature is <30°, then do dorsal plication.
5. If >30°, then divide the urethral plate.
6. If still >30°, then do ventral corporal lengthening (usually staged).

- Mild is <30°, moderate 30–45° and severe ≥45°. Eyeballing underestimates curvature.
- Plication risks nerve injury, recurrence and shortening. Lengthening uses corporotomies, with or without a dermis or tunica vaginalis graft.
- Recurrence is 28% after plication vs 9% after grafting, and not dividing the plate is an independent risk, so correct fully the first time.

8 · URETHROPLASTY

Tubularised incised plate (TIP) repair is used for >90% of distal repairs.

Technique	Indication	Key point / pitfall
MAGPI	Glanular, mobile meatus, no curvature	Used more proximally gives a retrusive meatus
Urethromeatoplasty	Stenotic distal meatus	Heineke–Mikulicz
GAP (glans approximation)	Wide deep groove (MIP)	Two-layer neourethra
Thiersch–Duplay	Wide plate	Tubularise over 12 Fr
TIP / Snodgrass	Distal, midshaft; narrow plate; mild curvature	Midline relaxing incision; stop 2–3 mm short; meatal stenosis; inlay graft if poor plate
Mathieu (flip-flap)	Distal, shallow plate	Narrow base causes ischaemia, fistula; fewer stenoses than TIP
Onlay island flap	Plate kept but narrow	Diverticulum if too wide; less stenosis
Duckett tube	Proximal; plate divided	Higher complications; diverticulum
Two-stage Bracka	Proximal severe curvature, redo, poor plate	Stage 1 quilted preputial/buccal graft; stage 2 (≥6 months) tubularise
Two-stage Byars	Proximal severe curvature	Dorsal prepuce flaps; highest staged complications

- Staged repair (STAG) has the lowest proximal complication rate (≈ 23%), because urethroplasty over a fresh corporotomy or graft fails. Repeat the artificial erection at stage 2. Proximal TIP has been abandoned because of devascularisation and recurrent curvature.

> **KEY POINT**
>
> Why proximal repairs fail (Poiseuille): resistance rises with length / radius⁴. A long, rigid tube ending in a non-distensible glans raises proximal pressure, causing fistula, diverticulum and stricture. 25–75% need extra procedures after planned two-stage repair.

- Glansplasty needs a tension-free **two-layer closure**, and a **barrier layer** (dartos or tunica vaginalis) reduces fistula.

9 · POST-OPERATIVE CARE

- Leave a compressive dressing for ≈ **72 h**, and divert urine for **3–21 days** according to complexity. Short distal repairs need no stent. Give oxybutynin until the stent is out, with paracetamol and ibuprofen.

10 · COMPLICATIONS

Complication	Standard definition
Fistula	Urethral leak below meatus
Glans dehiscence	Wing separation; glans fusion <2 mm
Meatal stenosis	Obstructive symptoms and **<8 Fr** prepubertal / <12 Fr postpubertal
Diverticulum	Segmental sacculation on voiding
Recurrent curvature	**>30°** on erection

P4 *Hypospadias: present concepts in management, and hypospadias fistula — commonest complication (<10%); distal obstruction key; wait 6–12 months; calibrate, stress test, layered closure with barrier; coronal fistula needs redo urethroplasty; recurrence ≈ 20%*

Not in the paper answer:

- **Balanitis xerotica obliterans** needs complete excision and a staged graft. Recurrence is up to **40%**.
- **Meatal stenosis** is commoner after TIP than Mathieu and is treated by dilatation, then meatotomy or redo. A late **stricture** needs buccal urethroplasty, often staged.

> **KEY POINT**
>
> Nearly every complication traces to distal obstruction, ischaemia or tension, so repair begins by calibrating the distal urethra.

Reoperation

- Excise scar widely and transfer tissue rather than doing "small operations".
- **Buccal mucosa** is the graft of choice. Make it 10% larger, then tubularise ≥6 months later; lip mucosa suits the glans. Always do cystoscopy and an artificial erection.
- The **Cecil–Culp** procedure buries the penis in the scrotum when skin is short.

11 · OUTCOMES AND FOLLOW-UP

- A TIP meta-analysis (4,675 boys) found complications in 10.6% and reoperation in 4.5%. Distal success is 83–95%, while proximal complications are 23–68%.
- About 75% of complications occur by 3 years, so follow proximal cases **through puberty**. Recurrent UTI needs imaging for diverticulum, stricture or utricle.
- Proximal cases have worse adult sexual outcomes, and penile length is the chief complaint.

57 Etiology, Diagnosis and Management of the Undescended Testis

1 · DEFINITIONS

A retractile testis is watched; an undescended one is operated on.

P1 P3 *Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis; Maldescended (undescended) testes and treatment algorithm — undescended vs ectopic (SIP commonest), retractile, acquired (8–11 years), vanishing (monorchia/anorchia) vs agenesis*

Term	Key point (beyond the papers)
Undescended testis (UDT)	Not in the normal scrotal position (midpoint at or below mid-scrotum). It is palpable or non-palpable; non-palpable means undescended or absent.
High scrotal testis	Stable, above the scrotal midpoint; a mixed group, not routinely called undescended.
Retractile	One that rarely stays down or rests high may progress to acquired UDT.
Recurrent	Undescended at birth, descended spontaneously, then re-ascended (10–22%, at 1–5 years).
Secondary / testicular retraction	Suprascrotal after hernia repair / after orchidopexy (iatrogenic).
Agenesis	Never formed; comes with ipsilateral Müllerian persistence (extremely rare).

2 · EMBRYOLOGY OF DESCENT

P1 *Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis —*
SRY/SF1, then SOX9, then Sertoli; INSL3 via RXFP2 + androgen; gubernaculum attaches to epididymis; transabdominal 10–15 weeks (outgrowth, CSL regression); scrotum by DHT at 10–12 weeks

Not in the paper answer:

- The gubernaculum does not pull the testis; it swells, then regresses.

Phase	Timing	Event	Control
Transabdominal (relative)	10–15 weeks	Gubernacular swelling (outgrowth) from 12 weeks; cranial mesonephric ligament regresses by 13 weeks	INSL3 and androgen act together
Inguinoscrotal	**20–28 weeks**	The canal widens and the processus forms; rapid passage through the canal; the gubernaculum then regresses. Most are scrotal after 27 weeks.	Androgen via the genitofemoral nerve (**GFN**), which releases calcitonin gene-related peptide (**CGRP**)

> **KEY POINT**
> Outgrowth creates the canal; regression lets the testis follow, and a defect in either gives the same phenotype. INSL3 and androgen act on one target (gubernacular/cremaster myogenesis), hence one clinical picture whichever fails.

Minipuberty is a hormonal surge peaking at 1–3 months, when gonocytes become **adult dark (Ad) spermatogonia**, the stem-cell pool. This is where a positional defect becomes a germ cell deficit.

3 · EPIDEMIOLOGY, AETIOLOGY AND ASSOCIATIONS

Spontaneous descent ends by about 6 months.

- Prevalence is **2–4% at birth** (term), falling to **1–2% at 3 months**; up to 45% in preterm boys.

- About 88% of high scrotal testes descend spontaneously, but only about half of extrascrotal ones.
- 80–85% are isolated. Risk factors are **prematurity and low birth weight**, family history, maternal diabetes and smoking. Diethylstilbestrol doubles risk.
- It is a polygenic threshold trait, and part of **testicular dysgenesis syndrome** with hypospadias, infertility and testicular germ cell tumour (TGCT).
- No hormone predicts germ cell number, so hormones are not tested routinely outside DSD and anorchia.

Associated condition	Cryptorchidism	Implication
Prune-belly syndrome	All (bilateral abdominal)	Abdominal orchidopexy
Spigelian hernia	53–80%	Absent gubernaculum/canal
Cerebral palsy	25–54%	Often bilateral
Gastroschisis	5–25%	**50–55% descend spontaneously — observe**
Omphalocele, imperforate anus, myelomeningocele, posterior urethral valves	12–33%	Caudal-field anomalies are ipsilateral in 90%
Congenital diaphragmatic hernia	7–30%	Always ipsilateral
Klinefelter (47,XXY)	> 50% of prepubertal Klinefelter boys	< 2% of non-syndromic UDT; more with hypospadias
Persistent Müllerian duct syndrome (PMDS)	Bilateral 59%; hernia uteri inguinalis; transverse ectopia 22%	AMH or AMHR2 mutation; adult infertility ≥ 80%, malignancy risk

Congenital vs acquired

Feature	Congenital	Acquired
Diagnosis	Birth/neonatal	Mean 8–11 years
Position	Often higher	Lower; often superficial inguinal pouch (SIP)
Processus vaginalis	Patent (87% unilateral)	Usually closed (patent 45–50%)
Epididymal anomalies	37–60%	11–31%

An acquired UDT is a never-fully-descended testis unmasked by growth, with semen results like congenital cases. Retractile testes share this pathology and ascend in about 2–3%, so they need **annual examination through childhood**.

4 · HISTOLOGY, ASSOCIATED ANOMALIES AND FERTILITY

Germ cells are lost from late infancy, so operate early.

- Germ cell loss begins **after infancy** and progresses; the contralateral testis is affected less.
- The UDT is smaller at birth, so the defect is intrinsic and worsened by position.
- Spermatogonia per tubule (S/T) is 2.5 at birth, with a nadir of 1.2 at 2–3 years.
- A **patent processus** is present in 87% of unilateral and 71% of bilateral cases. A UDT is at higher torsion risk.

Anomaly	Key features	Management
Polyorchidism	Usually left, scrotal in 75%; Kumar classification by drainage (A with vas, B without)	Observe a normal scrotal testis; orchidopexy if undescended with intact drainage
Splenogonadal fusion	Left side; the continuous form comes with limb defects and micrognathia	Recognise it to avoid a needless orchidectomy

Anomaly	Key features	Management
Transverse testicular ectopia	Hernia with a contralateral non-palpable testis; PMDS in 20–50%	Laparoscopy; trans-septal placement if vasa fused

Trial	Design	Result	Take-home
Kollin et al.	Randomised trial: orchidopexy at 9 months vs 3 years	Early: more Sertoli and germ cells, better testicular growth (to 4 years)	Operate in first year; age at surgery, not hormones, predicts germ cells

Fertility outcome

Outcome	Former unilateral	Former bilateral
Oligospermia / azoospermia	31% / 14%	42% / 31%
Paternity	**≈ 90% of those attempting — as controls**	≈ 65% of those attempting (≈ 1/3 overall)

- In unilateral cases the normal testis carries fertility, so semen analysis overstates infertility; in bilateral cases it reflects true infertility.
- Ad spermatogonia at biopsy predict adult sperm counts; if they are absent, all counts are subnormal.
- Even after first-year surgery, 20–25% remain at infertility risk. Surgery **before 18 months** reduced assisted reproduction, and each 6-month delay added 5%.

5 · MALIGNANCY

Orchidopexy lowers cancer risk only partly.

- TGCT risk is **2–5×** (pooled odds ratio (OR) ≈ 4). About 1% of cryptorchid men get TGCT, and 5–10% of TGCT patients had a UDT.
- In the **Swedish population study** the OR was about 2.8 if treated before puberty vs about 5.0 after. † Pettersson 2007: relative risk (RR) 2.23 if operated before 13 years vs 5.40 at 13 or older. A meta-analysis gave RR **5.8** for orchidopexy after 10–11 years.
- Untreated adults have OR ≈ 14.
- A persistently cryptorchid testis develops **seminoma** (74%), whereas tumours after orchidopexy are mostly non-seminoma (63%).
- Germ cell neoplasia in situ occurs in 2–3% of formerly cryptorchid men, so biopsy is for DSD and chromosomal cases. Biopsy at orchidopexy does not raise TGCT risk.
- Postpubertal UDT with a normal contralateral testis needs **orchidectomy** (up to age 50), with a prosthesis at the same sitting.
- **Microlithiasis** is not an independent TGCT risk, so routine ultrasound surveillance is not needed.

6 · DIAGNOSIS

Examination decides; imaging adds nothing.

Examination — the gold standard

ALGORITHM

1. Warm room and hands. Examine supine, then cross-legged or squatting.
2. With lubricated fingers, sweep one hand from the anterior superior iliac spine (ASIS) along the canal and palpate the lowest position with the other.
3. Compress the canal and invaginate the scrotum to feel the SIP.
4. Apply sustained traction to fatigue the cremaster. If it **stays, it is retractile; if it springs back, it is undescended**.

- About **80% palpable**, and 60–70% are unilateral. At surgery a non-palpable testis is abdominal or peeping in 25–50%, vanished in 15–40% and a missed extra-abdominal testis in 10–30%.
- Examination under anaesthesia (EUA) is the final arbiter.

P3 **P1** *Maldescended (undescended) testes and treatment algorithm; Vanishing testes — AUA algorithm:*

bilateral non-palpable, then DSD; re-examine at 6 months corrected age; EUA, then yearly review / orchidopexy / exploration / laparoscopy; contralateral hypertrophy > 2 mL (specificity 100%) suggests monorchia, explore anyway

Imaging — not recommended

The AUA (2014) advises no ultrasound or MRI. Ultrasound sensitivity is only about 45%, and imaging merely delays surgery.

Bilateral non-palpable or UDT with hypospadias: DSD work-up

- A newborn with bilateral non-palpable testes needs an **urgent karyotype and hormones** to exclude **congenital adrenal hyperplasia (CAH)**, because of salt-wasting.
- Proximal hypospadias with UDT carries a chromosomal abnormality in 32–47% (see Ch 56).
- Micropenis with bilateral UDT suggests hypogonadotrophic hypogonadism or anorchia.

Marker	Anorchia	Bilateral abdominal UDT
FSH, LH	Raised	Normal
AMH, INHB (best prepubertal Sertoli markers)	Undetectable/low	Present
Basal testosterone	Low in first year only	Normal
Testosterone after hCG	No rise	Rises

hCG stimulation is no longer the test of choice, and **laparoscopy remains the confirmatory test**.

7 · MANAGEMENT PRINCIPLES AND TIMING

- **Timing**: observe to **6 months** (corrected age), then refer; operate at **6–18 months**, preferably by 12.
- Retractile testes need no surgery but yearly examination. Acquired UDT should be operated on, especially if bilateral.
- A pubertal or postpubertal boy with a normal contralateral testis needs orchidectomy, with or without a prosthesis.

Hormonal therapy — not recommended (AUA)

- Intranasal LHRH gives descent in only about 20%, and hCG was never placebo-tested. hCG may increase germ cell apoptosis and reduce adult volume.
- Adjuvant LHRH for bilateral UDT with absent Ad spermatogonia (Hadziselimovic) has pilot data only.

> **KEY POINT**
>
> Hormones fail on both claims for one reason: position, not endocrinology, is the modifiable variable. Descent is not androgen-limited after fetal life, and germ cells track age at scrotal placement.

8 · SURGERY FOR THE PALPABLE TESTIS

The goal is a tension-free scrotal testis with intact vessels.

Inguinal orchidopexy — key steps

ALGORITHM

1. EUA, skin-crease incision, check the SIP.
2. Open the external oblique, sparing the ilioinguinal nerve.
3. Free the testis; divide the gubernaculum distal to the sac (protect a long-looping vas).
4. **Separate the patent processus from vas and vessels without skeletonising; high-ligate it at the internal ring** (main length gain).
5. If short, divide lateral spermatic bands and dissect retroperitoneally; then the **Prentiss manoeuvre** (medial to the inferior epigastric vessels).
6. Transverse scrotal incision and **subdartos pouch**; pass the testis untwisted.
7. Close dartos around the cord; transalbugineal sutures are unnecessary.

- If the testis cannot reach the scrotum, stage the procedure. Orchidectomy is only for an atrophic or abnormal testis, a postpubertal boy or an insufficient vas.
- Biopsy is not routine; it is indicated in DSD and considered in bilateral UDT or after 12–13 years.

Scrotal (Bianchi) orchidopexy

The testis is delivered through a high scrotal incision, the processus ligated and the testis fixed in a subdartos pouch. It is increasingly first choice for palpable, especially low, testes.

9 · THE NON-PALPABLE TESTIS

Laparoscopic findings at the ring decide the next step.

ALGORITHM

1. EUA: if now palpable, do an inguinal or scrotal orchidopexy.
2. If **still non-palpable, do diagnostic laparoscopy** (ring patency, vessels, vas, testis).
3. If **blind-ending vessels** (with or without vas) lie above a closed ring, the testis has vanished: stop (the only unequivocal endpoint).
4. If **vessels and vas enter the ring**, explore the groin/scrotum and excise the nubbin (viable tubules 5–15%).
5. If the testis is **intra-abdominal**, do a primary laparoscopic or Fowler–Stephens (one-or two-stage) orchidopexy; orchidectomy if postpubertal or dysgenetic.
6. If it is peeping at the ring, do a laparoscopic or inguinal orchidopexy.
7. If neither vas nor vessels are seen, search the perivesical area and retroperitoneum to the kidney.

- An open ring usually, but not always, means a viable distal testis, and apparently blind-ending vessels may coexist with an ectopic testis, so keep exploring if equivocal.
- The AUA does not require fixing the contralateral solitary testis.

Laparoscopic orchidopexy

ALGORITHM

1. Empty bladder and stomach. Open (Hasson) umbilical port; CO_2 at 8–12 mmHg. Inspect both rings and add working ports.
2. Divide the gubernaculum distal to the ring and use it as a handle.
3. Incise peritoneum lateral to the vessels and distal to the vas; mobilise the vessels, **preserving the peritoneal triangle between vas and vessels** (keeps Fowler–Stephens possible).
4. Make a new hiatus medial to the inferior epigastric vessels.
5. Bring the testis through a trans-scrotal port to a subdartos pouch without tension.

- Ring closure is unnecessary, and the robot offers no clear advantage.

Fowler–Stephens (FS) and alternatives

P1 *Methods of orchidopexy — FS on vasal/gubernacular collaterals for testes not within 2–4 cm of ring; success table: primary laparoscopic > 90–97%, Jones/Bagley ≈ 95%, FS 60–97% (one-stage 63%, two-stage 68%), Shehata 84–85% (atrophy 10%), autotransplant 96%; Dessanti PTFE traction*

Not in the paper answer:

- Two-stage FS: clip the vessels high, wait about **6 months**, then mobilise on a wide peritoneal pedicle. Two-stage is slightly favoured.
- Keeping peritoneum over the vasal vessels and **sparing the gubernaculum** lowers atrophy. Avoid dividing vessels when possible, because it reduces germ cells.
- Operate on bilateral high testes about 6 months apart.

> **KEY POINT**
>
> Every abdominal technique answers one question: how to gain length without losing blood supply. FS sacrifices the testicular artery; traction gains length over time while keeping it, so vessel division is the last resort.

10 · COMPLICATIONS AND FOLLOW-UP

Complication	Cause / prevention	Management
Atrophy (≈ 2% modern; historically 24% abdominal/peeping vs 10% distal)	Vessel injury, skeletonisation, FS, tension	Offer a **prosthesis** ≥ 6 months after scrotal surgery or after puberty, inguinal route, fixed to dartos
Retraction / non-scrotal (≈ 2–10%)	Inadequate retroperitoneal mobilisation, unligated processus	Redo: a prescrotal testis by the scrotal approach; otherwise a lateral approach through unscarred tissue or a fascial patch on the cord. Results equal primary surgery.
Vas injury	Long-looping vas cut with the gubernaculum	Divide the gubernaculum distal to the sac; avoid skeletonising
Torsion after orchidopexy	Rare	Prevented by subdartos fixation
Laparoscopic	Bladder/vessel injury, hypercapnia, bowel obstruction	Hasson entry, bladder drainage

Follow-up: examine at ≥ **6 months** for position and size. Counsel on fertility and cancer, and teach self-examination from puberty.

61 Urologic Considerations in Pediatric Renal Transplantation

P1 *Urological evaluation and management of pediatric kidney transplant patient. — causes, five goals, pre-transplant screen (VCUG/urodynamics indications), bladder optimisation and cycling, augment segments, transplant technique, complication rates and management*

Not in the paper answer:

- Graft reflux matters more than native reflux, because the kidney is solitary and immunosuppressed.

1 · CAUSES OF ESKD IN CHILDREN

- The commonest cause of end-stage kidney disease (ESKD) is congenital anomalies of the kidney and urinary tract (**CAKUT**), then obstructive uropathy, mainly posterior urethral valves (**PUV**).
- Older children have FSGS or neuropathic bladder. The commonest age at transplant is **13–17 years**.

KEY POINT

Five goals: unimpeded drainage · capacious low-pressure reservoir with social continence · volitional emptying · no infection · fewest operations. Chronic rejection is unavoidable, but every added insult accelerates it.

2 · PRE-TRANSPLANT EVALUATION

- Every child needs a history and renal ultrasound, plus uroflow with post-void residual if toilet-trained.
- VCUG and urodynamics are selective: neuropathic bladder, PUV, reflux, prior surgery, febrile UTI or worsening hydronephrosis.

ALGORITHM

1. Positive history: US and VCUG. If dynamics are abnormal, urodynamics.
2. Treat with drugs and clean intermittent catheterisation (CIC), with or without bladder cycling, then reassess.
3. If still unsafe, augment; then clear for transplant.

3 · BLADDER MANAGEMENT

The bladder must be safe before the graft goes in.

- The commonest pattern is **low capacity, overactivity and poor compliance**, and high pressure harms the kidney even without reflux.
- The **target** is expected capacity at detrusor pressure **< 40 cm H_2O**, with social continence.
- Use antimuscarinics, mirabegron or botulinum toxin, with CIC as needed.
- Cycle a **defunctionalised bladder** with rising saline volumes, then repeat urodynamics. **Never use the allograft to test a bladder.**
- **Augment** if the bladder cannot reach > **75%** of expected capacity at < 40 cm H_2O with CIC no more often than every **3–4 h** despite maximal therapy. Postpone this in PUV, but augment earlier in neuropathic bladder.
- A dilated ureter is the best segment. Bowel augments need lifelong malignancy surveillance.
- Site a **Mitrofanoff medial to the planned transplant incision**.
- Reconstruct **before** transplant, allowing **6–12 weeks** to heal, except when transplanting before 2 years.

4 · NATIVE NEPHRECTOMY

P2 *Pre transplant nephrectomy — default keep; indications (hypertension, nephrotic, polyuria, infection/VUR, stones, malignancy, size); retroperitoneal on PD, embolisation if high risk; leave normal*

ureter, remove dilated refluxing ureter

Not in the paper answer:

- Combine it with the transplant only to make room, as in a small child with huge kidneys.

5 · TRANSPLANT INTO THE ABNORMAL BLADDER

- A Gibson retroperitoneal incision is standard; a child < **15 kg** may need the aorta and vena cava through a transperitoneal approach.
- The preferred join is an **extravesical non-refluxing ureteroneocystostomy**. **Ureteroureterostomy** to the native ureter is the alternative and the salvage.
- Stent if difficult, removing the double-J at **4–6 weeks**.

6 · UROLOGICAL COMPLICATIONS

Complication	Rate	Timing / management
Ureteric stenosis	**6%**	Most within 100 days; distal, ischaemic. Focal: balloon and stent for 4–6 weeks; otherwise open repair (native ureter, bladder flap); do not delay.
Urine leak	**3%**	Early. Drain-fluid creatinine; check catheter, then US, then MAG3 scan or cystogram. Small: observe; large: drain or explore. Short necrosis: reimplant; long: native ureter, psoas hitch or flap.
Significant VUR	2%	See below
Stones	2%	Treat if > 1 cm and growing, obstructing or infected; one-stage modality

Rising creatinine with hydronephrosis

ALGORITHM

1. Indwelling catheter for 1–2 weeks: if creatinine falls, the bladder is the cause.
2. If not, stent and keep the catheter; biopsy if rejection is suspected.
3. If retrograde stenting fails, nephrostomy and antegrade study.

- The graft is unusually sensitive to obstruction, and obstruction and rejection coexist, so stent at a low threshold.
- An **allograft mass is post-transplant lymphoproliferative disorder (PTLD) until proven otherwise**.
- Recurrent febrile UTI from graft reflux with an optimised bladder needs an **open or robotic reimplant**; asymptomatic reflux is observed.

7 · OUTCOMES

- Urological complications rarely lose the graft, but **≥ 2 UTIs** lower graft GFR at 1 and 4 years.

63 Paediatric Urologic Oncology: Renal and Adrenal

1 · WILMS TUMOUR (WT): EPIDEMIOLOGY AND GENETICS

Genetics decide who is screened and who needs kidney-sparing surgery.

P2 *Wilms tumour: classification, staging and management — genes (WT1 11p13, IGF2 11p15), syndrome risks, screening, anaplasia, SIOP risk groups, COG staging, COG vs SIOP, nodes, adjuvant by stage*

Not in the paper answer:

- WT is **6–7%** of childhood cancers and **95%** of kidney cancers under 15 y; median age **3.5 y**, younger if bilateral or syndromic. Refer for genetics if bilateral, under 2 y, with rests, syndromic features or family history.
- WAGR (Wilms, aniridia, genitourinary anomalies, intellectual disability) is a WT1 deletion; **Denys–Drash** syndrome (DDS) is a **dominant-negative** point mutation with early end-stage renal disease. Type I tumours (WT1) are stromal with intralobar rests; Type II (11p15, Beckwith–Wiedemann syndrome, BWS) are blastemal/epithelial with perilobar rests.
- Combined 1p and 16q loss of heterozygosity (~5%) earns the intensified **regimen M**; 1q gain (25–30%) is the next stratifier.

Syndrome risk	Examples
High (> 20%)	WAGR, DDS (**50%**); Fanconi D1 (BRCA2); Perlman (DIS3L2); mosaic variegated aneuploidy
Moderate (5–10%)	BWS, isolated hemihypertrophy, Frasier (WT1), Simpson–Golabi–Behmel (GPC3)

Syndrome risk	Examples
Low	Li–Fraumeni, neurofibromatosis 1, Sotos, trisomy 18, Bloom

KEY POINT

Trap: BWS is the **commonest** predisposing syndrome but only moderate risk (21% bilateral); WAGR/DDS are rarer and highest.

ALGORITHM

1. If WT risk is > 1%, do a renal ultrasound (US) every **3–4 months** until **7 y**.
2. If a lesion looks suspicious, do computed tomography (CT) or magnetic resonance imaging (MRI).

- In BWS, hemihypertrophy and similar syndromes, add α-fetoprotein (AFP) 3-monthly to 4 y for hepatoblastoma.
- Screening lowers stage and allows nephron-sparing surgery (NSS), but survival benefit is unproven.

2 · PATHOLOGY

- Favourable histology is **triphasic**; the tumour is friable and spills easily.
- **Anaplasia** (chemoresistant): nuclei ≥ **3×** neighbours, hyperchromasia and abnormal mitoses; it is commoner in older children (~13% over 5 y).
- Nephrogenic rests are found in > 1/3 of WT kidneys and **> 90%** of bilateral cases; most regress. Perilobar rests are peripheral (BWS); intralobar rests lie anywhere (WAGR/DDS).

3 · CLINICAL FEATURES AND WORK-UP

- A varicocele persisting supine, or hepatomegaly, suggests inferior vena cava (IVC) thrombus. Gross haematuria needs cystoscopy at surgery.
- **Acquired von Willebrand disease** occurs in up to 8%; give desmopressin.
- Ascites beyond the cul-de-sac best predicts rupture; operate urgently only for bleeding. Imaging cannot judge nodes or separate WT from renal cell carcinoma (RCC) or clear cell sarcoma (CCSK).

4 · STAGING AND COG VS SIOP

COG operates first; SIOP gives chemotherapy first.

- Children's Oncology Group (COG) stages I–V are surgical–pathological. Stage III means **positive nodes, any spill**, implants, a positive margin or piecemeal removal.
- The International Society of Paediatric Oncology (SIOP) gives pre-operative vincristine (VCR) and dactinomycin (AMD) without biopsy.

KEY POINT

COG stage is surgical-pathological, so the surgeon, not the scan, decides chemotherapy intensity and irradiation: one spill converts stage I/II to III.

5 · SURGICAL PRINCIPLES

- Do a transperitoneal radical nephrectomy and inspect the liver, peritoneum and nodes.
- Sample nodes (COG **≥5 nodes**); omission is the commonest protocol violation and under-stages the tumour.
- Spill (~10%) causes local relapse; complications (~11%) are mainly haemorrhage and small-bowel obstruction.
- COG favourable histology: stage I–II get VCR and AMD; stage III–IV add doxorubicin (DOX) with flank/abdominal (and lung) radiotherapy (RT). Cyclophosphamide and etoposide are added in regimen M and for CCSK.

Bilateral WT (stage V) and NSS

P3 *Bilateral Wilms tumour: treatment strategies — 5–7%; no upfront nephrectomy/biopsy; 3 drugs 6 weeks, then NSS by week 12 (NWTS-4 anaplasia); open bilateral biopsy (discordant 4/5); enucleation + flank RT; transplant after 1–2 y; AREN0534 EFS 82%, OS 95%*

Not in the paper answer:

- Use frozen section and a normal-kidney margin in NSS.

ALGORITHM

1. Synchronous bilateral WT: **no upfront nephrectomy, no biopsy**.
2. Give VCR, AMD and DOX for 6 weeks, then re-image.
3. If NSS is feasible, operate.
4. If not, give more chemotherapy, but do **surgery by week 12**.
5. If the tumour does not respond, do a **bilateral open biopsy**.

- Unilateral NSS suits syndromic children or a solitary kidney; UMBRELLA criteria are **<300 mL**, an adequate remnant and negative nodes.

IVC thrombus

- It occurs in ~6–10%. Below the hepatic veins, do upfront nephrectomy with **en bloc** thrombectomy (piecemeal means stage III).
- **At or above the hepatic veins, give pre-operative chemotherapy** to avoid bypass.

6 · RELAPSE AND LATE EFFECTS

Relapse group	Definition	Salvage
Standard	Non-anaplastic, previously VCR/AMD only	70–80%
High	Non-anaplastic, after ≥ 3 drugs	40–50%
Very high	Anaplastic/blastemal	~10%

- Late effects track **RT and anthracycline**: heart failure follows DOX (rare below **150 mg/m²**). Renal failure after unilateral nephrectomy is only **0.6%** at 20 y.

7 · OTHER RENAL TUMOURS

Tumour	Key facts	Management
Congenital mesoblastic nephroma (CMN)	Commonest **neonatal** renal tumour; ETV6–NTRK3 fusion in the cellular type	Radical nephrectomy alone; adjuvant only if incompletely resected
Clear cell sarcoma (CCSK)	3%; **bone** and brain metastases; late relapse; never bilateral	DOX-containing chemotherapy; RT even at stage I
Rhabdoid tumour (RTK)	2%; **most lethal**; SMARCB1/INI1 loss; associated brain tumours	No standard; tazemetostat on trial
Cystic nephroma / cystic partially differentiated nephroblastoma (CPDN)	Identical on imaging; CPDN has blastema in the septa; DICER1	Excision cures
RCC	Outnumbers WT above 13–14 y. **Translocation RCC (TFE3/TFEB, Xp11.2)** is commonest.	Complete resection; resistant to chemotherapy and RT
Angiomyolipoma	Tuberous sclerosis; bleeding risk if > 3 cm and growing	Mammalian target of rapamycin (mTOR) inhibitor first; embolise for bleeding

KEY POINT

Age points to diagnosis: a neonate has CMN, a toddler WT or rhabdoid, an adolescent RCC. Metastatic site sets staging tests: lung (WT), bone (CCSK), brain (rhabdoid/CCSK).

8 · NEUROBLASTOMA

Biology, not radical surgery, decides outcome.

- It is the commonest **extracranial solid tumour** and commonest infant malignancy; median age is 19 months, 50% are adrenal and ~70% are metastatic at diagnosis.
- Paraneoplastic syndromes include watery diarrhoea from vasoactive intestinal peptide and **opsoclonus–myoclonus–ataxia**.
- Urinary vanillylmandelic acid and homovanillic acid (**VMA/HVA**) are raised in **90–95%**. **Calcification and vessel encasement** favour neuroblastoma over WT.
- 123I-meta-iodobenzylguanidine (MIBG) scanning is required for staging; MRI is best for intraspinal extension. Screening caused **overdiagnosis** without fewer deaths.

International Neuroblastoma Risk Group (INRG) stage	Definition
L1	Localised, **no** image-defined risk factor (IDRF)
L2	Localised, ≥ 1 IDRF (vessel/nerve encasement)
M	Distant metastases
MS	Age **<18 months**; skin, liver, marrow (< 10%) only; no bone (INSS 4S: < 12 months)

- INRG is imaging-based, before treatment; the International Neuroblastoma Staging System (INSS) depends on resection and is surgeon-dependent.
- **MYCN amplification** (> 10 copies; ~20%) means high risk at any stage. M at ≥ 18 months is always high risk.
- Adverse features are 1p36 deletion, 11q LOH, 17q gain and **diploidy**; favourable features are **hyperdiploidy**, TrkA expression and age < 18 months.

KEY POINT

Trap: diploid is bad, hyperdiploid good. Stage 4S/MS has OS 80–88% with frequent regression; infants < 2 months may die of massive hepatomegaly or coagulopathy.

- **Low risk**: resection alone (stage 1 > 90% disease-free); chemotherapy only for symptoms, progression or adverse biology. Never remove the kidney or vital structures.
- **4S**: resecting the primary is not mandatory. **Intermediate**: chemotherapy, then resection.
- **High risk**: induction chemotherapy, delayed surgery after 3–4 courses, myeloablative consolidation with autologous stem-cell rescue, RT 15–30 Gy, and **13-cis-retinoic acid with anti-GD2 antibody**.
- In stage 4, radical resection improves local control but not survival.
- **Cord compression**: give chemotherapy first; laminectomy only for progressive deficit, as it causes scoliosis.

KEY POINT

Surgical value runs inversely with biology: favourable tumours are cured by little or no resection; unfavourable ones are not rescued by radical resection, so vital structures are never traded for margin.

9 · ADRENOCORTICAL TUMOURS

P4 *Adrenal mass evaluation and adrenocortical carcinoma: diagnosis and management — bimodal (children < 10 y, 90% functional); Li–Fraumeni/BWS; open en bloc, intact capsule; mitotane + EDP*

Not in the paper answer:

- Offer **TP53** testing to every child. Androgen excess causing **virilisation** is the commonest presentation.
- Work-up includes DHEA-S, testosterone, 17-hydroxyprogesterone, cortisol tests, a **urinary steroid profile**, metanephrines, CT/MRI and chest CT.
- Weight > 200 g and spill are adverse. Give **perioperative steroids** for cortisol-secreting tumours.

> **KEY POINT**
> Why open? Cure needs complete removal of an intact tumour; capsular rupture seeds the retroperitoneum, with poor salvage.

10 · LANDMARK TRIALS

Trial	Design	Result	Take-home
AREN0532/0533	1p/16q LOH treated with regimen M	Stage III–IV event-free survival (EFS) ~90% vs 61%	Biology-driven intensification
AREN0533 (lung)	Lung nodule response at week 6	RT omitted, EFS and overall survival (OS) preserved	Response-adapted RT omission
AREN0534	Bilateral WT: 3-drug, surgery by week 12	EFS 82%, OS 95% (vs 56%/81% in NWTS-5)	Fixed clock, maximise NSS
COG P9641	Low-risk neuroblastoma, surgery alone	EFS ~89%, OS ~97%	Chemo only if symptomatic
COG anti-GD2	Anti-GD2 + cytokines + retinoid vs retinoid	Antibody superior	Immunotherapy standard in high risk
German NB97	Extent of resection, stage 4 > 18 months	No EFS/OS benefit	Radical resection not essential

64 Paediatric Urological Oncology: Pelvis, Bladder, Testis and Genitalia

1 · RHABDOMYOSARCOMA (RMS): BIOLOGY

Fusion status, not the microscope, now sets risk.

- RMS is the **commonest childhood sarcoma**; 20–30% are genitourinary (GU), at bladder/prostate (BP), paratesticular (PT) or vulvovaginal (VV) sites. GU-RMS 5-y survival is **>80%**.
- Syndromic cases (Li–Fraumeni, DICER1, neurofibromatosis 1, Costello) are usually embryonal; diagnosis under 3 y should prompt a germline search.

	Embryonal (ERMS)	Alveolar (ARMS)
Site	GU (commonest); the **botryoid** variant protrudes from bladder or vagina	Trunk, extremities, perineum
Age	Younger (~70% < 10 y)	Older, adolescents
Fusion	Usually negative	~80% **PAX3/7–FOXO1** [t(2;13), t(1;13)]
Prognosis	Better	Fusion-positive worse; fusion-negative ARMS behaves like ERMS

KEY POINT

Fusion status predicts outcome better than histology, so trials assign risk by **fusion status**.

2 · PRESENTATION AND WORK-UP

- BP tumours cause haematuria or retention; PT tumours are a painless mass separate from the testis; VV tumours cause bloody discharge or a prolapsing mass.

ALGORITHM

1. Start with US.
2. CT or MRI of the primary and its nodal basin: BP, pelvic and retroperitoneal nodes up to the renal vessels; PT, retroperitoneal; VV, inguinal, pelvic and retroperitoneal.
3. **Chest CT** (lung is the commonest metastatic site).
4. Bone scan and bilateral marrow sampling; omit these in fusion-negative ERMS without nodal or lung disease.

3 · STAGE, GROUP AND RISK

The **stage** is pretreatment tumour–node–metastasis plus site. The **clinical group** records what the surgeon left. The **risk group** adds fusion status and age.

Children's Oncology Group (COG) stage	Site	Size / nodes
1	Favourable: **paratesticular, female GU**, orbit, head–neck	Any, M0
2	Unfavourable: **bladder/prostate**, extremity, parameningeal, trunk	≤ 5 cm, N0/NX
3	Unfavourable	≤ 5 cm N1, or > 5 cm any N
4	Any	M1

N1 means a node ≥ 1 cm in short axis. BP-RMS is never below stage 2.

Clinical group	Definition
I	Localised, completely resected, nodes negative
IIa / IIb / IIc	IIa: microscopic residual. IIb: involved nodes resected, most distal node negative. IIc: microscopic residual or most distal node positive.
III	Gross residual or **biopsy only**
IV	Distant metastases
Risk	Criteria
Low	Fusion-negative; stage 1 group I–II (III orbit only), or stage 2 group I–II
Intermediate	Fusion-negative stage 1–2 group III (non-orbit), stage 3 group I–II; any fusion-positive non-metastatic; stage 4 fusion-negative < 10 y
High	Stage 4 fusion-positive, or ≥ 10 y

KEY POINT

An organ-preserving **biopsy** places BP-RMS in group III, which means **intermediate risk**; PT-RMS (resected by orchidectomy) is almost always low risk. Age < 1 y or > 10 y is independently adverse. Trap: most distal node positive = IIc, not IIb.

4 · TREATMENT PRINCIPLES

Chemotherapy does most of the work; surgery aims to save the organ.

- **Chemotherapy is universal**; local control is by radiotherapy (RT) and/or surgery. Complete resection (except PT) is done only if the organ is preserved.
- Primary exenteration fell from almost all cases in IRS-I to **13%** in IRS-IV, with overall survival unchanged.
- **Pretreatment re-excision** (before chemotherapy, without organ loss) improves GU survival and can change the group. **Delayed primary excision** cannot change the group and is done only if a negative-margin resection is likely.

Bladder/prostate RMS

- Biopsy is preferably **cystoscopic**; relieve obstruction with a **urethral catheter** first and avoid a suprapubic tube (seeding).
- An **end-of-therapy residual mass** is usually stroma or mature rhabdomyoblasts, so observe it if stable. Viable tumour after chemoradiation may need cystectomy.
- Only ~40% of preserved bladders are "normal", so do **urodynamics at 6 months** and renal US yearly for ≥ 10 y.

Paratesticular RMS

- Do a **radical inguinal orchidectomy**; after a scrotal approach, excise the remaining cord inguinally.
- Do an **ipsilateral nerve-sparing retroperitoneal lymph node dissection (RPLND)** before chemotherapy if **age ≥10 y** or nodes are radiologically N1.

> **KEY POINT**
> Why age, not CT? Older boys have far more retroperitoneal disease and CT has a high false-negative rate for nodes.

Female GU, perineal, infants

- Vaginoscopy and biopsy; chemotherapy with or without brachytherapy controls most, and exenteration is reserved for unresponsive disease.
- Perineal tumours are mostly fusion-positive alveolar, with 5-y OS ~50%. RT causes most late effects, including bladder and sexual dysfunction, infertility and second cancers.

5 · OTHER BLADDER TUMOURS

- **Urothelial** tumours in children are mostly low grade; treat by transurethral resection and follow with US. A recurrence should prompt a search for a syndrome (Lynch).
- **Spina bifida/exstrophy**: lifetime bladder cancer risk is **2–4%**, aggressive, and **not increased by augmentation**; evaluate haematuria promptly.

Benign lesion	Key point	Treatment
Inflammatory myofibroblastic tumour	ALK rearrangement; intermediate malignancy	Complete resection
Nephrogenic adenoma	Follows trauma, infection, surgery or transplant	TURBT and treat the cause; recurs
Eosinophilic cystitis	Biopsy essential; blood eosinophilia often absent	Steroids, antihistamines
Haemangioma	Severe haematuria; Klippel–Trenaunay, Sturge–Weber	Propranolol, laser; partial cystectomy

6 · TESTICULAR TUMOURS

Prepubertal tumours are mostly benign, so the testis can often be spared.

- Before puberty, **teratoma** is the commonest tumour (benign) and **yolk sac tumour (YST)** the commonest malignant one.
- Work-up is Doppler US with LDH, **α-fetoprotein (AFP)** and β-hCG; an **"onion-skin" appearance suggests epidermoid cyst**.

	AFP	**β-hCG**
Source	**YST invariably**	Embryonal carcinoma, mixed tumours; rare before puberty
Half-life	**5 days**	**24 h**
Caveat	Normally high in infancy; reaches adult level (< 10 ng/mL) by **6–9 months**	—

ALGORITHM

1. Use an inguinal approach.
2. If markers are normal and US looks benign, do a **testis-sparing partial orchidectomy** with frozen section.
3. If benign, close.
4. If frozen section shows germ cell malignancy, or AFP is raised, do a radical inguinal orchidectomy.

- Prepubertal teratoma needs partial orchidectomy and **no follow-up**; postpubertal teratoma is malignant and needs radical orchidectomy.
- **YST**: radical inguinal orchidectomy; **90%** are stage I, managed by **surveillance without RPLND or chemotherapy**. Stage III–IV disease gets cisplatin, etoposide and bleomycin.
- **Stromal** tumours (Leydig, Sertoli, juvenile granulosa) are treated by partial orchidectomy if markers are normal. Testicular adrenal rests in congenital adrenal hyperplasia respond to more steroid.

Disorders of sex development (DSD)

Tumour risk rises with Y material in dysgenetic gonads: high in gonadal dysgenesis, Frasier, Denys–Drash and partial androgen insensitivity (PAIS); intermediate in complete androgen insensitivity (CAIS); low without Y. Gonadoblastoma is premalignant.

> **KEY POINT**
>
> Trap: **complete** androgen insensitivity is intermediate (not high) risk; partial is high.

7 · OTHER PELVIC AND GENITAL TUMOURS

- **Sacrococcygeal teratoma** causes neurogenic bladder in 36–51%, so follow with renal–bladder US and urodynamics for higher types. The **Currarino triad** also causes neurogenic bladder.
- A Sertoli–Leydig ovarian tumour should prompt **DICER1 screening**.
- **Urethral prolapse** mimics RMS in prepubertal girls; treat with topical oestrogen.

★ Putting it together: common paediatric decisions

A · VUR AFTER FEBRILE UTI

ALGORITHM

1. Do renal–bladder US; add a voiding cystourethrogram (VCUG) if UTIs recur or US is abnormal.
2. Treat bladder and bowel dysfunction (BBD) and constipation first.
3. Low risk (grade I–II, older boy) is observed; high risk (grade ≥ III, girl, BBD, scars, infant) gets continuous antibiotic prophylaxis.
4. Breakthrough UTI or new scars: inject or reimplant (see Ch 41, 53).

B · DILATED UPPER TRACT

Condition	Key test	Operate when	Operation
Pelviureteric junction (PUJ) obstruction	MAG3 diuretic renogram	Function falling or < 40%, symptoms	Anderson–Hynes pyeloplasty
Primary megaureter	US, MAG3, VCUG	Falling function, infection; most resolve	Tapered reimplant
Ureterocele / ectopic ureter	US, VCUG, functional scan	Obstruction, infection, incontinence	Incision, heminephrectomy or ureteroureterostomy (see Ch 52)
Multicystic dysplastic kidney	US and DMSA scan	Rarely; most involute	Observe

C · UNDESCENDED TESTIS

ALGORITHM

1. **Palpable**: orchidopexy at **6–18 months**; no imaging.
2. **Non-palpable**: examination under anaesthesia, then laparoscopy to find and fix the testis or confirm it is absent.
3. **Bilateral non-palpable, or with hypospadias**: DSD work-up first (see Ch 59).

D · ACUTE SCROTUM

Never let a scan delay exploring torsion.

	Testicular torsion	Appendage torsion	Epididymitis
Typical	Sudden pain, high-riding testis	Upper-pole tenderness, "blue dot"	Gradual onset
Action	**Explore at once**; fix both sides, remove if non-viable	Analgesia, rest	Culture; antibiotics if infected

E · NEWBORN WITH ATYPICAL GENITALIA

KEY POINT

Bilateral impalpable gonads with atypical genitalia = salt-wasting congenital adrenal hyperplasia until proven otherwise. Check electrolytes, 17-hydroxyprogesterone and karyotype; crisis comes at days 5–7.

F · WILMS TUMOUR VS NEUROBLASTOMA

	Wilms tumour	Neuroblastoma
Age	Median 3.5 years	Median ~18 months
Imaging	Intrarenal claw sign; IVC extension	Calcification, vessel encasement; crosses the midline
Markers	WT1, 11p15; WAGR, Denys–Drash	VMA/HVA raised in 90–95%; MYCN amplification = high risk

	Wilms tumour	Neuroblastoma
Strategy	COG: nephrectomy first; SIOP: chemotherapy first	By INRG risk group